



ACQLERATE

MODULE 4

Data Analytics for Program Managers

Use data to drive decisions, measure performance, and communicate program health with confidence.

8 Lessons · ~174 Minutes · Lesson Book Edition

Course Lessons

01	Program Metrics & KPIs	13 min
02	Data-Driven Decisions & Visualization	12 min
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DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 1

Program Metrics & KPIs

13 min

Identify and track the key performance indicators that matter most in defense program management.

Your Program's Vital Signs

Think of program metrics the way a doctor thinks about vital signs. Blood pressure, heart rate, temperature: each tells you something specific about what's happening and what might go wrong. A good PM has a small set of metrics that do the same for their program: cost burn rate, schedule variance, technical progress, risk burn-down. The goal isn't to collect data -- it's to see problems coming before they become visible to everyone else. The PM who walks into a program review already knowing what's wrong, why it's wrong, and what they're doing about it is the PM leadership trusts.

Live Example: Reading a Program's Health

This program has spent \$9M of budget and completed \$8M of planned work on a \$20M contract. Here's what the numbers say:

- **CPI: Cost Performance Index** Getting \$0.89 of work per \$1 spent — $EV \div AC = \$8M \div \$9M = 0.89$
- **SPI: Schedule Performance Index** 80% of planned work complete — $EV \div PV = \$8M \div \$10M = 0.8$
- **TCPI: To-Complete Performance Index** Must earn \$1.17 per \$1 to hit budget — $(BAC - EV) \div (BAC - AC) = \$12M \div \$11M = 1.17$

CPI < 0.90 and SPI < 0.90 means this program is both over budget and behind schedule. TCPI > 1.10 means recovery to original budget is unrealistic: EAC needs to be updated.

Core PM Dashboard Metrics

CATEGORY	METRIC	WHAT IT TELLS YOU	WARNING SIGN
Cost	CPI (EV/AC)	Cost efficiency of work performed	CPI < 0.90
Schedule	SPI (EV/PV)	Schedule efficiency vs. plan	SPI < 0.90
Cost	VAC (BAC - EAC)	Projected overrun/underrun at completion	VAC < -5%
Technical	TPM achievement %	Technical parameter vs. planned maturity	< 80% on schedule
Risk	Open high-risk items	Count of unmitigated high risks	> 5 high items
Quality	Defect closure rate	Rate of defect identification vs. resolution	Backlog growing
Schedule	Critical path float	Days of slack on program critical path	< 10 days float

TECHNICAL PERFORMANCE MEASURES (TPMS)

TPMs are contractually defined parameters with threshold (minimum acceptable) and objective (desired) values, tracked against a planned maturity profile over the program timeline. For a communications system, a TPM might be "data throughput $\geq$ 100 Mbps." If the contractor's current throughput is below the planned maturity curve, the PM knows: before integration testing; that a technical risk is materializing.

Leading vs. Lagging Indicators

- **Leading (predictive): Staffing levels, test asset availability, GFE/GFI delivery status, risk burn-down rate, supplier delivery performance** Leading indicators let you intervene before problems become cost and schedule variances. If staffing is at 70% of plan in month 3, you have a problem that will show up in the EVM data in month 5: but only if you're tracking staffing now. GFE/GFI delays are another classic leading indicator: if the government is late delivering test equipment, the schedule impact arrives weeks later.
- **Lagging (historical): CPI, SPI, defect count, actual vs. planned milestones, budget execution rate** CPI and SPI are the most common lagging indicators: they tell you what already happened. A CPI of 0.87 means 13 cents of cost overrun for every dollar of work accomplished, confirmed. This is valuable diagnostic information but it describes the past. Leading indicators are what let you change the future.
- **Best practice: Track both: lead indicators give you time to act; lagging indicators confirm trends** The most effective program dashboards show both: leading indicators on the left (staffing, risk burn-down, GFE/GFI status) and lagging metrics on the right (CPI trend, SPI, milestone performance). This combination gives the PM a complete picture: what's happening now and what the trajectory suggests.
- **Red flag: If ALL your metrics are lagging, you're managing by looking in the rear-view mirror** Programs that report only CPI, SPI, and milestone completion are essentially confirming problems after they occur. By the time a CPI of 0.85 appears in the data, the root cause events happened 60-90 days earlier. A PM with only lagging indicators is always reacting, never anticipating.

THE DASHBOARD DESIGN RULE

A PM's dashboard should fit on one page and answer three questions at a glance: Are we on cost? Are we on schedule? Are we achieving technical performance? Any metric that doesn't answer one of those three questions is overhead noise: useful in deep dives but not in an executive dashboard.

INTERMEDIATE**Designing a Meaningful KPI Framework for Your Program**

Mid-career PMs often inherit a metrics framework designed by someone else. Or no framework at all. Building a good one requires discipline: limit leading indicators to 5-7 (more creates noise), ensure each KPI has a data owner responsible for accuracy, and tie thresholds to contract performance criteria not arbitrary round numbers. The most important distinction: leading indicators (staffing ramp, BCWP/BCWS trend, open action item age) predict future performance; lagging indicators (CPI, SPI, schedule slippage) confirm what already happened. Programs that only track lagging indicators are always reacting, never preventing. Build at least two leading indicators per major risk area. Review them weekly at the PM level, not just monthly at program reviews.

Thresholds That Mean Something

Most inherited dashboards use round numbers: CPI below 0.95 is yellow, below 0.90 is red. Nobody remembers where those came from and nobody acts on them. A threshold earns its place when it is tied to a consequence in your contract or your schedule. CPI below 0.92 matters if that is the point where your funded ceiling runs out before the period of performance ends. Critical path float below 10 days matters if your test range booking is 10 days out. Work backward from the thing that breaks: what value of this metric means I lose the range slot, blow the funding line, or miss the delivery? That number is your threshold, and because you can explain it, people act on it.

Where a Defensible Threshold Comes From

METRIC	WEAK THRESHOLD	THRESHOLD TIED TO A CONSEQUENCE
CPI	Below 0.95 is yellow	Below the value where projected EAC exceeds the funded ceiling
Critical path float	Below 10 days	Below the lead time of the next unmovable external event
Staffing fill rate	Below 90 percent	Below the level that misses the integration start date
Supplier on-time delivery	Below 95 percent	Below the rate that consumes your assembly buffer
Open high risks	More than 5	Any risk whose realized cost exceeds management reserve
Defect closure	Backlog growing	Backlog growth rate that pushes test completion past the milestone

The Review Rhythm: Who Looks at What, and How Often

Metrics without a cadence become a monthly slide nobody reads. The working pattern on healthy programs has three loops. Weekly, the PM and leads look at the leading indicators only: staffing, supplier deliveries, test asset availability, open action age. These change fast enough to act on. Monthly, the full set comes together with the contractor's cost report, and the question is whether the trend changed, not what this month's number was. Quarterly, you look at the baseline itself: is the plan we are measuring against still the plan? Each loop has a different owner and a different decision. If your weekly review is looking at cumulative CPI, you are reviewing history at a frequency that cannot change it.

Every Metric Needs Four Things Before It Goes on the Dashboard

- **A source of record** The system the number comes from, named. Not a spreadsheet someone maintains by hand, and not two systems that disagree.
- **An owner** A person accountable for the number being right and current, on the government side, not just the contractor's analyst.
- **A refresh cadence** How often it updates and how stale it can be before it is misleading. A number refreshed monthly cannot support a weekly decision.
- **A defined action** What happens when it crosses the threshold, and who does it. A threshold with no action is decoration.

ADVANCED**METRICS MANIPULATION: HOW CONTRACTORS GAME KPIS AND HOW TO CATCH IT**

Experienced senior PMs know that any metric a contractor is evaluated on will eventually be gamed. The classic EVM manipulations: retroactive replanning (moving budget into periods where work was already done), schedule logic manipulation (adding constraints to remove negative float), and loading budget into early periods then underspending to generate positive CPI artificially. For technical metrics: systems that report "green" on readiness metrics by narrowing the denominator (counting only tested units, not all units required). The countermeasure: build cross-referencing requirements into your metrics framework. If CPI is 1.05 but cost-per-unit is rising, something is wrong. If schedule shows green but critical path items keep slipping, the schedule logic is suspect. Require contractors to explain anomalies. Not just report metrics.

Building a Leading Indicator Set for Your Actual Risks

Generic dashboards track generic things. A senior PM builds the leading set from the program's own top risks, because a leading indicator is only leading if it predicts the specific failure you are worried about. If your top risk is software integration, the leading indicators are requirements churn rate, build failure rate, and defect discovery rate per build, none of which appear on a standard EVM dashboard. If your top risk is a sole-source supplier, the leading indicators are that supplier's on-time delivery, their staffing on your line, and their own supplier issues. Two leading indicators per top risk is the working rule. Any more and the dashboard stops fitting on a page; any fewer and you are watching one thermometer in a building with several fires.

Matching Leading Indicators to the Risk They Predict

PROGRAM RISK	LEADING INDICATORS THAT ACTUALLY PREDICT IT	LAG BEFORE IT SHOWS IN EVM
Software integration failure	Requirements churn, build failure rate, defect discovery per build	Two to four months
Sole-source supplier slip	Supplier on-time delivery, their staffing on your line, their own supplier alerts	One to three months
Staffing shortfall	Requisitions open over 60 days, clearance queue age, attrition rate	Two to three months
Test failure cascade	Test asset availability, dry-run pass rate, range schedule confirmations	One to two months
Requirements instability	Change requests opened per month, unresolved interface issues	Three to six months

When Your Metrics Disagree, Believe the Physical One

The most useful moments in metrics work are the contradictions. CPI is 1.03 but the cost per delivered unit keeps climbing. Schedule shows green while the critical path slips a week each month. Defect closure looks healthy while the same defects keep reopening. In every case one number is measuring the accounting system and the other is measuring the physical work, and the physical one is right. Cost efficiency can look fine while unit cost rises because the work being performed shifted to easier tasks. Schedule can hold while float erodes because the logic was changed. Build the habit of pairing each financial metric with a physical one: CPI with unit cost, SPI with critical path float, defect closure with reopen rate. When the pair disagrees, you have found something worth a meeting.

What the PEO Is Watching, and Why It Is Not Your Dashboard

Your program dashboard answers whether you are executing. The level above you is answering a different question: whether this program still belongs in the portfolio. That means unit cost against the baseline, the distance to a reportable breach, whether the schedule still supports the fielding date the requirement promised, and how much of the funding line you will actually obligate this year. A PM who only ever reports execution metrics gets surprised when the portfolio question arrives. The fix is to keep a second, four-number view for leadership and to put your own program's distance-to-breach on it before anyone asks. The lifecycle module covers what a baseline breach triggers; the point here is that the number should never be new to you.

WHAT SENIOR PMS DO DIFFERENTLY

They tie every threshold to a consequence they can name, so the dashboard produces decisions instead of colors. They build leading indicators from their own top risks rather than from a template. They pair every financial metric with a physical one and chase the contradictions. And they track their distance to a reportable breach themselves, so the portfolio conversation is one they start rather than one that happens to them.

One sentence to teach your team: **a metric earns its place on the dashboard only when it has a source of record, an owner, a refresh cadence, and a defined action at a threshold you can trace to something that actually breaks.**

BUILD ON THIS

- **Building a One-Page Program Dashboard** Turning this into the artifact you actually brief
- **Forecasting Burn** Where leading indicators become a prediction
- **Program Reviews and the Baseline** The portfolio question your metrics have to answer

Key Terms

KPI

Key Performance Indicator. A measurable value demonstrating how effectively an objective is being achieved.

Technical Performance Measure (TPM)

A parameter that tracks achievement of key technical requirements against a planned profile.

Threshold

The minimum acceptable value for a performance parameter; program fails if not met.

Objective

The desired best value for a performance parameter; program tries to achieve this.

Leading Indicator

A metric that predicts future performance (e.g., staffing levels predict schedule performance).

Lagging Indicator

A metric that reports past performance (e.g., CPI shows what has already happened).

Source of Record

The one system a metric is drawn from, named and agreed, so two briefings cannot show different numbers for the same thing.

Data Lineage

The traceable path from a number on the dashboard back to the transaction or measurement that produced it, including how stale it is.

Variance at Completion (VAC)

Budget at Completion minus Estimate at Completion. The projected overrun or underrun in dollars when the work is finished.

Critical Path Float

Days of slack on the longest chain of dependent tasks. When it reaches zero, every further slip moves the program's end date.

Knowledge Check

A Technical Performance Measure (TPM) with a "threshold" value represents:

- A. The ideal best-case technical outcome
- ✓ **B. The minimum acceptable technical performance: below this the program fails**
- C. The baseline value at contract award
- D. The contractor's internal target for performance

WHY

In DoD acquisitions, threshold values are the minimum acceptable performance levels for Key Performance Parameters (KPPs) and TPMs. If a system cannot achieve the threshold, it fails to meet the requirement. The "objective" value is the desired best value. PMs track both to understand performance margin.

Which of the following is a LEADING indicator of future schedule performance?

- A. SPI (Schedule Performance Index)
- B. Actual vs. planned milestone completion dates
- ✓ **C. Contractor staffing levels and key personnel fill rate**
- D. Cost variance percentage

WHY

Staffing levels are a leading indicator: understaffed programs have not yet shown schedule slippage in the SPI, but the lag in hiring predicts future schedule problems. SPI and milestone data are lagging indicators: they report what has already happened. Leading indicators are more valuable because they allow preventive action.

Critical path float of less than 10 days on a defense program indicates:

A. The program is ahead of schedule

✓ **B. There is minimal schedule buffer, and any delay on critical path activities will directly slip the milestone**

C. The program has sufficient margin for risk events

D. Float only matters on commercial programs, not defense

WHY

Critical path float (also called "slack") represents the amount of time an activity can be delayed without delaying the program's critical path milestone. Less than 10 days of float means the program has virtually no schedule buffer: any delay, vendor issue, or test failure that affects the critical path will directly slip the contract delivery date.

VAC (Variance at Completion) is calculated as:

A. EV minus AC

✓ **B. BAC minus EAC**

C. PV minus EV

D. CPI minus SPI

WHY

$VAC = BAC - EAC$. A negative VAC indicates a projected cost overrun at completion. For example, if $BAC = \$100M$ and $EAC = \$115M$, then $VAC = -\$15M$, meaning the program is projected to overrun by \$15M. VAC translates the current performance indices into a projected dollar impact at program completion.

A defense PM's dashboard shows CPI = 0.88, SPI = 0.92, and 8 open high-risk items. Based on standard DoD benchmarks, this program should be rated:

A. Green: all indicators are above 0.85

✓ **B. Yellow/Red: CPI below 0.90 is a red indicator requiring corrective action report**

C. Yellow: SPI is acceptable but needs monitoring

D. Green: risk count below 10 is acceptable

WHY

$CPI < 0.90$ is a "Red" indicator on standard DoD program dashboards, requiring a formal corrective action report. SPI of 0.92 is "Yellow." The combination of Red cost performance and 8 open high risks makes this a program requiring immediate attention. PMs should never rationalize away a CPI below 0.90.

The primary reason to include BOTH leading and lagging indicators in a program dashboard is:

A. To satisfy audit requirements

✓ B. Leading indicators enable preventive action; lagging indicators confirm whether interventions worked

C. They are required by DFARS for all ACAT I programs

D. To compare contractor performance against industry benchmarks

WHY

Leading indicators (predictive) allow the PM to see problems forming and take action before they appear in cost/schedule data. Lagging indicators (historical) confirm trends and validate whether corrective actions are working. Using only lagging indicators means managing reactively: always behind the problem.

For a defense program's executive dashboard, what is the recommended design principle?

A. Include every available metric to demonstrate thoroughness

✓ B. Fit on one page, answering: Are we on cost? On schedule? Achieving technical performance?

C. Focus exclusively on financial metrics since that is what leadership cares about

D. Use a separate dashboard for each program stakeholder

WHY

Executive dashboards should be concise and decision-focused. A well-designed PM dashboard fits on a single page and answers the three critical questions: cost performance, schedule performance, and technical achievement. Excessive detail belongs in backup briefings, not the executive summary.

A growing defect backlog (more defects being found than closed) is a warning sign indicating:

A. The test program is more thorough than expected: a positive sign

✓ B. Quality and schedule risk: the program is accumulating technical debt that will affect later milestones

C. Normal behavior during early integration testing

D. The contractor is ahead on testing, finding issues early

WHY

A growing defect backlog indicates the program is generating quality issues faster than it can resolve them: a strong predictor of test phase overruns and potential milestone delays. Even if defects are "expected" in early integration, a sustained or growing backlog is a risk that requires active management and root cause analysis.

Supplier delivery performance tracking is best classified as which type of indicator?

A. Lagging: it reports what suppliers have already delivered

✓ **B. Leading: delays in supplier deliveries predict future schedule problems for the prime contractor**

C. Technical performance measure

D. A financial indicator only

WHY

Supplier delivery performance is a leading indicator. When a critical component supplier starts slipping delivery dates, the prime contractor's schedule will eventually be impacted: but often weeks or months later. Tracking supplier on-time delivery gives PMs early warning to identify alternative sources or accelerate efforts before the prime program schedule is affected.

TPMs (Technical Performance Measures) are tracked against a "planned maturity profile" because:

A. All technical parameters must be 100% achieved at contract award

✓ **B. Technical performance is expected to improve progressively; the profile shows whether achievement is on pace**

C. DFARS requires a specific TPM format at every program review

D. TPMs are only meaningful at final delivery testing

WHY

TPMs are not expected to achieve their final values immediately: they mature progressively as development proceeds. The planned maturity profile shows what value should be achievable at each point in time. Tracking actual TPM values against the profile tells the PM whether technical development is proceeding on pace or falling behind: before the program reaches system integration testing.

Order the steps for putting a new metric on your program dashboard:**CORRECT ORDER**

1. Name the decision the metric is supposed to inform
2. Identify the consequence that defines the threshold
3. Name the source of record and the person who owns the number
4. Set the refresh cadence against how fast the decision must be made
5. Write down the action that fires when the threshold is crossed

WHY

Name the decision first. A metric with no decision behind it is decoration, however good the data is.

Match each financial metric to the physical metric that should be paired with it:

CORRECT MATCHES

CPI	Cost per delivered unit
SPI	Critical path float
Defect closure rate	Defect reopen rate
Obligation rate	Work physically completed

WHY

Pairing an accounting measure with a physical one is how you catch the contradictions that reveal real problems.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 2

Data-Driven Decisions & Visualization

12 min

Use charts, dashboards, and data storytelling to communicate program status and drive decisions.

Data Tells a Story: Make Sure It's the Right One

Numbers without context mislead. A \$10M cost overrun means very different things on a \$100M contract vs. a \$10B contract. Effective program managers present data in ways that correctly convey the program's health status: not to hide problems, not to exaggerate them, but to enable the right decision at the right level of leadership. Misleading data presentation is a leadership failure and an integrity issue.

Essential Charts for a PM's Toolkit

- **S-Curve: Cumulative BCWS vs. BCWP vs. ACWP: the fundamental EVM visualization. Any divergence between the three lines tells the whole cost/schedule story.** The S-Curve gets its name from the characteristic shape of cumulative spending on a program: slow start, acceleration through EMD, plateau in production. BCWS (Budget Cost Work Scheduled) is the plan line. BCWP (Budget Cost Work Performed) is actual progress. ACWP (Actual Cost Work Performed) is actual spending. BCWP below BCWS = schedule slip. ACWP above BCWP = cost overrun.
- **Gantt/IMS Bar Chart: Timeline view of all program activities, dependencies, and critical path. Updated monthly.** The Integrated Master Schedule is the Gantt chart on steroids: it shows every work package, milestone, and dependency relationship. The critical path is the longest path through the network from start to finish; any delay on the critical path delays the program end date by the same amount. Weekly schedule scrubs should focus on critical path tasks first.
- **Risk Matrix (5×5): Plot all risks by probability and impact. Shows at a glance where attention is needed.** The 5×5 risk matrix scores each risk from 1-5 on probability and 1-5 on impact, creating a 25-cell grid. High-probability/high-impact risks in the upper right require immediate mitigation action and management reserve. The matrix should be updated monthly and briefed at every program review: a static risk matrix is a sign of a program that isn't actively managing risk.
- **Burn Rate Chart: Monthly and cumulative obligation vs. plan. Reveals execution patterns.** Burn rate (obligation rate) is closely watched by Congressional staffers and OSD budget reviewers. Programs that obligate significantly below plan face "use it or lose it" pressure at fiscal year end: or lose funding in the next POM cycle. Programs that overoblite early may have cash flow problems later. The burn rate chart shows both and makes the pattern visible.
- **TPM Trend Chart: Actual technical parameter values vs. planned maturity profile over time.** Technical Performance Measures (TPMs) track how actual system performance is maturing relative to planned development. If the radar range KPP calls for 300km detection at CDR and you're at 180km with 6 months to

go, the TPM chart shows that trajectory clearly. TPMs are the early warning system for technical risk that will eventually become cost and schedule risk.

- **Waterfall/Variance Chart: Decompose cost variance by work package or contractor to identify root causes.** Variance analysis is the "so what" of EVM data. A \$5M negative cost variance is useless information without knowing which work packages are driving it, why, and what the PM is doing about it. The waterfall chart breaks the total variance into its components: showing, for example, that 80% of the cost overrun is in the propulsion subsystem due to a specific technical challenge.

THE S-CURVE: YOUR PROGRAM'S EKG

An S-Curve plots BCWS (planned), BCWP (earned), and ACWP (actual) on the same axis over time. The gap between BCWP and ACWP is cost variance. The gap between BCWP and BCWS is schedule variance. Any experienced acquisition professional can read a program's entire cost/schedule health from a properly formatted S-Curve in 30 seconds.

DATA PRESENTATION FOR EXECUTIVES

Executive briefings demand different data presentation than program team reviews. For executives: lead with the conclusion (not the data), use traffic light color coding (red/yellow/green), limit to 3 charts maximum, and have backup data ready for questions. For program teams: show all the detail: waterfall variances, WBS-level trends, and root cause analysis.

Chart Selection Guide

WHAT YOU WANT TO SHOW	BEST CHART TYPE
Overall cost & schedule health over time	S-Curve (BCWS/BCWP/ACWP)
Where cost variance is coming from	Waterfall chart by WBS element
Program schedule and dependencies	Gantt / IMS bar chart
Risk portfolio status	5×5 Risk matrix
Budget execution vs. plan by month	Bar chart (planned vs. actual burn)
Technical parameter trend vs. target	Line chart with threshold/objective bands
Comparing multiple programs	Radar/spider chart or normalized bar chart

INTERMEDIATE

Presenting Data to Non-Technical Decision Makers

Mid-career PMs spend a significant portion of their time briefing flag officers, SES officials, and congressional staff who have limited technical background. The most effective briefings for this audience: one chart per major message, direct headline titles that state the conclusion (not "CPI Trend" but "Cost Performance Is Recovering. CPI Up 0.08 in 90 Days"), and explicit "so what" statements. Avoid EVM jargon in flag-level briefings. Translate: "negative SV" becomes "we are behind schedule"; "VAC" becomes "projected overrun at completion." Spend 80% of your preparation on the narrative, 20% on the charts. Decision makers who can't understand your data can't support your program.

One Chart, One Decision

The fastest way to improve a program briefing is to give every chart a job. Before you build it, write the sentence the chart is supposed to make true: "we will run out of funded ceiling in March unless we descope" or "the supplier, not our team, is driving the slip." If you cannot write that sentence, the chart is decoration and it is costing you attention you will need later. This also tells you when a chart is finished. A burn chart that shows obligations against a funded line, with the crossing point marked, has done its job. The same chart with three extra series, a secondary axis and the full work breakdown has not done a better job; it has done four jobs badly.

The Chart Failures That Show Up in Every Program Review

- **Dual axes** Two different units on left and right make any two lines appear related. Split them into two stacked charts that share an x-axis instead.
- **Truncated y-axis** Starting the axis at 94 percent turns a two-point move into a cliff. Start at zero unless you say plainly that you did not.
- **Pie charts with more than four slices** Nobody can rank the middle slices. A sorted bar chart answers the same question in one glance.
- **Rainbow categories** More than about six colors and the legend becomes a lookup table. Gray everything except the one or two series that carry the point.
- **The unlabeled threshold** A chart with no line for the number that matters makes the reader do the comparison in their head, which they will do wrong.
- **Month-to-month noise** Plotting a volatile monthly figure without its trend invites the room to argue about the last data point instead of the direction.

Annotate the Chart, Do Not Narrate It

A chart that needs you standing next to it is a chart that fails the moment your slide is forwarded to someone who was not in the room, which is how most decisions actually get made. Put the finding in the title: not "Monthly Obligations," but "Obligations trail plan by \$2.1M; funded through March." Mark the one point that matters with a short label on the chart itself. Keep the legend out of the way, or drop it entirely by labeling the lines where they end. The test is simple: hand the page to someone unfamiliar with the program and ask what it says. Whatever they tell you is what the chart says, regardless of what you meant.

The Five-Chart Monthly Review Set

CHART	QUESTION IT ANSWERS	DECISION IT SUPPORTS
Funding burn against funded ceiling	When do we run out of money?	Reprogram, descope, or request funds now
Cost and schedule index trend, twelve months	Is performance getting better or worse?	Intervene, or stop intervening
Critical path float trend	Is the end date still real?	Re-sequence, add capacity, or renegotiate the date
Staffing fill against plan	Can we do the work we promised?	Escalate hiring, or replan the ramp
Top risks with movement since last month	What is most likely to hurt us next?	Fund a mitigation, or accept and watch

ADVANCED

BUILDING A PROGRAM DATA ARCHITECTURE: FROM CONTRACT TO DASHBOARD

Senior PMs on large programs should architect the data flow from the start: IPMR data → government EVMS analysis tool (SMART, Cobra, wInsight) → program office dashboard → leadership briefing. Each translation introduces error and delay; minimize handoffs. Define your data schema before contract award: what format does the contractor submit IPMR data in? What are your visualization requirements? For software programs, define your agile metrics pipeline: how does sprint velocity, defect backlog, and code coverage data flow from contractor dev tools (Jira, GitLab) to your dashboard? Programs that build data architecture retroactively spend 6-12 months cleaning inconsistent data before they can make it useful. Contract for the data architecture. Not just the deliverable.

Show the Range, Not the Line

A single forecast line communicates a precision nobody has. The honest version of a forecast is a band: the pessimistic case if current performance holds, the plan case, and the optimistic case with stated

assumptions. Senior audiences respond well to this once they understand what it buys them, because the width of the band is itself information. A narrow band on a mature program says the estimate is stable. A band wide enough to cross the funded ceiling says the decision is not really about the midpoint, it is about whether you can live with the top of the range. When you are asked for one number anyway, give the number and the condition attached to it: "\$18.4M if the supplier holds their current delivery rate; \$19.6M if it stays where it has been for the last three months."

The Same Data, Two Audiences

The government team and the contractor are looking at the same cost report and drawing different pictures from it, and a senior PM knows what the other side's version looks like. The contractor's view emphasizes what is inside their control and what the government caused: growth traceable to direction, late government-furnished property, requirement changes. The government's view emphasizes performance against the negotiated baseline. Neither is dishonest. The practical move is to build the bridge chart yourself, the one that starts at the original baseline and walks through each change to the current estimate, categorized by who drove it. Bring that to the review before the argument happens, and the conversation moves from whose chart is right to what to do next.

Building the Pipeline So the Chart Is Never the Bottleneck

On a program with a monthly rhythm, the analyst rebuilding the deck by hand each month is a single point of failure and a source of quiet errors. The architecture worth investing in is boring: one source of record per number, an extraction that runs on a schedule rather than on request, a small set of stored views that the charts read from, and a versioned template so last month's chart and this month's are actually comparable. The payoff is not prettier charts. It is that when leadership asks an unscheduled question in week two, you can answer it that day, and the answer ties to the number you briefed in week one. The dashboard lesson in this module covers the artifact; this is the plumbing that keeps it honest.

WHAT SENIOR PMS DO DIFFERENTLY

They write the sentence the chart has to prove before they build it. They put the finding in the title so the page survives being forwarded. They show forecasts as ranges with the assumption attached rather than as a single confident line. They build the baseline-to-current bridge before the contractor does. And they invest in the pipeline so an unscheduled question does not cost a week.

One sentence to teach your team: **every chart should have one job, say its finding in the title, and show a threshold or a range, because a chart that needs you standing beside it fails the moment it is forwarded.**

BUILD ON THIS

- **Building a One-Page Program Dashboard** The artifact this discipline produces
- **Forecasting Burn** Where the range replaces the line
- **Stakeholder Management & Communication** Reading the room you are briefing

Key Terms**S-Curve**

A cumulative cost/schedule plot showing planned vs. actual expenditure over time. The standard EVM visualization.

Burn Rate

The rate at which a program is consuming its budget over time.

Waterfall Chart

Shows how an initial value is affected by positive and negative changes. Useful for variance analysis.

Trend Line

A line projected from historical data showing where the program is headed if current performance continues.

Data Normalization

Adjusting data to a common scale for meaningful comparison (e.g., cost per unit, not total cost).

Source of Record

The single system a number is drawn from, so two briefings cannot show different values for the same metric.

Confidence Band

The range around a forecast showing the pessimistic and optimistic cases with their assumptions, rather than a single line.

Bridge Chart

A chart that walks from the original baseline to the current estimate one change at a time, attributing each step to what caused it.

Small Multiples

A grid of small charts with identical axes, used to compare many series without stacking them into one unreadable plot.

Annotation

A label placed on the chart itself marking the point that matters, so the finding survives without the presenter.

Knowledge Check

An S-Curve for an EVM program plots which three curves?

- A. BAC, EAC, and VAC over time
- ✓ **B. BCWS (PV), BCWP (EV), and ACWP (AC) over time**
- C. CPI, SPI, and TCPI over time
- D. Planned staffing, actual staffing, and required staffing

WHY

The S-Curve plots three cumulative cost lines over time: BCWS (Planned Value: what was planned to be spent), BCWP (Earned Value: the budget value of work accomplished), and ACWP (Actual Cost: what was actually spent). The gaps between these three lines visually reveal cost variance (EV vs. AC) and schedule variance (EV vs. PV).

For an executive-level program briefing, which data presentation approach is most effective?

- A. Present all available data in detail to demonstrate transparency
- ✓ **B. Lead with the conclusion, use traffic-light color coding, limit to 3 charts, have backup detail ready**
- C. Present only positive results to maintain stakeholder confidence
- D. Use WBS-level waterfall charts as the primary visualization

WHY

Executive briefings require a different approach than working-level reviews. Executives need to make decisions quickly: they need the "so what" upfront, color-coded health indicators, and a limited number of charts. Detailed backup should be available for Q&A. Burying the status in raw data is a failure to communicate effectively.

A waterfall chart in program management is best used to:

A. Show cumulative cost trends over time

✓ **B. Decompose cost or schedule variance by WBS element to identify root causes**

C. Display the program schedule and critical path

D. Compare contractor staffing against requirements

WHY

A waterfall chart decomposes a total variance into its contributing elements, showing which WBS areas, contractors, or contract line items are driving cost or schedule problems. It allows a PM to answer the question "WHERE is the variance coming from?" rather than just knowing the total variance magnitude.

On an S-Curve, if the BCWP (EV) line is consistently below the BCWS (PV) line, this indicates:

A. The program is over budget

✓ **B. The program is behind schedule: less work has been accomplished than planned**

C. The program is technically deficient

D. Contractor staffing is insufficient

WHY

On an S-Curve, BCWP below BCWS represents a schedule variance ($SV = EV - PV < 0$): the program has accomplished less work than planned. This is a schedule problem, not necessarily a cost problem. If ACWP is also above BCWP, there is simultaneously a cost problem. Reading S-Curve gap patterns is a core EVM skill.

Data normalization is important when comparing program metrics because:

A. It reduces the volume of data to present

✓ **B. It adjusts data to a common scale, enabling meaningful comparison (e.g., cost per unit rather than total cost)**

C. It removes classified information from reports

D. It is required by DFARS for IPMR reporting

WHY

Normalization adjusts raw data to a common scale so meaningful comparisons can be made. A \$10M overrun on a \$50M program is far more significant than a \$10M overrun on a \$5B program: normalizing to percentage reveals the relative magnitude. Without normalization, comparing programs of different scales produces misleading conclusions.

The primary purpose of a 5×5 risk matrix in a program review is to:

- A. Calculate the exact dollar cost of each risk
- ✓ **B. Visually display all program risks by probability and impact to prioritize management attention**
- C. Replace the formal risk register with a simpler tool
- D. Satisfy DCMA surveillance requirements

WHY

A 5×5 risk matrix plots risks on a grid of probability (1-5) vs. impact (1-5), creating a visual "heat map" that immediately shows where management attention should focus: the high-probability, high-impact upper-right quadrant. It complements (but does not replace) the detailed risk register with a quick-look visualization for leadership.

A program's monthly burn rate chart shows actual obligations consistently 20% below planned. The most likely implication for future-year funding is:

- A. The program is performing efficiently and will save money
- ✓ **B. The program may face future-year funding reductions due to apparent under-execution**
- C. The program needs a contract modification to reduce the ceiling
- D. The government should immediately terminate the contract

WHY

Consistent under-execution signals to higher headquarters that the program either doesn't need as much money as it requested, or is unable to efficiently execute its current funding. This is the "use it or lose it" dynamic: programs that routinely under-execute face FYDP reductions, as budget analysts assume the excess funding can be applied to higher-priority programs.

A trend line projected from historical CPI data is valuable to a PM because:

A. It is the official DoD method for calculating EAC

✓ **B. It shows where program cost performance is heading if current trends continue, enabling proactive intervention**

C. It satisfies the IPMR Format 5 requirement

D. It replaces the need for contractor EAC submissions

WHY

Trend lines (often called "CPI trend analysis") extrapolate historical performance into the future, giving PMs a data-driven view of likely outcomes absent intervention. A declining CPI trend that is not yet in "red" territory but is consistently worsening should prompt investigation and corrective action before a formal threshold breach.

When presenting a program's TPM (Technical Performance Measure) data, including the "planned maturity profile" alongside actual values is important because:

A. It is required by DFARS 252.234-7002

✓ **B. It allows stakeholders to assess whether technical achievement is pacing correctly, not just whether the final target has been met**

C. The planned profile determines the contractor's award fee

D. It replaces the need for system testing data

WHY

The planned maturity profile contextualizes TPM data: a radar system achieving 60 Mbps throughput is either excellent or poor depending on whether the plan called for 40 Mbps or 80 Mbps at that point in development. Without the planned profile, decision-makers cannot assess whether technical development is on track.

A radar chart (spider chart) is most useful for which type of program management analysis?

A. Showing cumulative cost and schedule trends over time

✓ **B. Comparing multiple programs or contractors across several dimensions simultaneously**

C. Displaying the critical path and dependencies

D. Calculating EAC from CPI data

WHY

Radar/spider charts plot multiple dimensions (e.g., cost, schedule, technical, risk, quality) for multiple entities on the same chart, making them ideal for portfolio-level comparisons. A PEO reviewing 10 programs can quickly identify which programs are strong across all dimensions and which have specific weaknesses. They're not appropriate for single-program trend analysis over time.

Order the steps for building a chart that will survive being forwarded:**CORRECT ORDER**

1. Write the sentence the chart has to make true
2. Pick the chart form that answers that question most directly
3. Put the finding in the title rather than a generic label
4. Mark the threshold, the crossing point, or the range on the chart itself
5. Hand it to someone outside the program and ask what it says

WHY

Writing the finding first is what keeps a chart to one job, and putting it in the title is what makes it travel.

Match each chart failure to the fix:**CORRECT MATCHES**

Dual axes implying a relationship	Two stacked charts sharing an x-axis
Truncated y-axis exaggerating a move	Start at zero, or state plainly that you did not
Pie chart with seven slices	A sorted bar chart
Rainbow of series colors	Gray everything except the one or two that carry the point

WHY

Each of these shows up in most program reviews, and each has a standard remedy.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 3

EVM Acronym Deep Dive

30 min

Master every EVM acronym. BCWS, BCWP, ACWP, CPI, SPI, EAC, ETC, VAC, TCPI. With clear definitions, formulas, and PM-level context.

What EVM Actually Measures

EVM stands for Earned Value Management. At its core it is just a way of comparing three numbers: what you planned to spend, what you have actually accomplished in budget terms, and what you have actually spent. Every acronym in this lesson traces back to one of those three source numbers. Learn the three, and every formula built on top of them makes sense. EVM integrates three independent measurement systems, cost accounting, schedule management, and performance measurement, into one framework. Every acronym is a specific data point. Confusing EAC with ETC or VAC with SV is not just embarrassing in a program review: it leads to wrong decisions. All calculations flow from three source numbers: BCWS (planned), BCWP (accomplished), ACWP (spent).

The Three Source Numbers: Everything Starts Here

Every EVM metric is calculated from these three values. Know them cold.

BCWS / PV: Budgeted Cost of Work Scheduled (Planned Value)

BASELINE

What you planned to spend by today

The cumulative budget for all work planned to be completed as of the data date. Comes directly from the time-phased PMB.

- **Full Name** Budgeted Cost of Work Scheduled
- **Modern Equivalent** PV (Planned Value)
- **Source** Sum of work packages planned to start/complete through the data date per the approved schedule
- **What It Tells You** Where you should be on the S-curve. Denominator for $SPI = EV/PV$.

BCWS/PV does NOT tell you what was spent or what was done. Only what was supposed to be done. The power of EVM comes from comparing BCWS to BCWP (actual work done) and ACWP (actual dollars spent).

BCWP / EV: Budgeted Cost of Work Performed (Earned Value) EARNED VALUE

The heart of EVM: value of work actually completed

The budget VALUE of work actually completed as of the data date. The only metric that directly measures accomplishment: not spending.

- **Full Name** Budgeted Cost of Work Performed
- **Modern Equivalent** EV (Earned Value)
- **Formula** % Complete × Budget for that work package
- **Critical Distinction** BCWP is based on BUDGET, not actual cost. If a \$100K task is 50% complete, BCWP = \$50K regardless of whether you spent \$30K or \$90K.

Earning Methods matter: Discrete (milestone-based: 0/50/100, 25/75), Apportioned (tied to related work), Level of Effort (LOE: time-phased only, used for management overhead). LOE work packages always show SV=0 and never show behind-schedule.

ACWP / AC: Actual Cost of Work Performed (Actual Cost) ACTUAL COST

Real dollars spent on completed work

Actual dollars incurred to accomplish the work measured by BCWP. Comes from the accounting system.

- **Full Name** Actual Cost of Work Performed
- **Modern Equivalent** AC (Actual Cost)
- **Source** Accounting system actuals: payroll, accounts payable, subcontractor invoices, material receipts
- **Key Insight** If ACWP > BCWP you are spending more than the work is worth (overrun). If ACWP < BCWP you are under-running.

Variances: How Far Off Are You?

Variances tell you the dollar magnitude of cost and schedule problems.

CV: Cost Variance CV = BCWP – ACWP

Over or under budget for completed work?

Negative CV = spending more than the work is worth. Positive CV = under budget.

- **Formula** CV = BCWP – ACWP (EV – AC)
- **Positive CV** Under budget. Green.
- **Negative CV** Over budget. Red. Requires Variance Analysis Report (VAR).
- **Related Index** CPI = BCWP / ACWP. CPI > 1.0 = efficient. CPI < 1.0 = over budget.

SV: Schedule Variance $SV = BCWP - BCWS$

Ahead or behind schedule in dollar terms?

Negative SV = less work completed than planned. Measured in dollars, not days.

- **Formula** $SV = BCWP - BCWS$ ($EV - PV$)
- **Positive SV** Ahead of schedule. Green.
- **Negative SV** Behind schedule. Red.
- **Limitation** SV converges to zero at contract completion regardless of how late you finish. Use Earned Schedule (ES) for late-phase schedule analysis.

VAC: Variance at Completion $VAC = BAC - EAC$

Projected total overrun or underrun

VAC tells you whether the program will finish within budget. This is the number executives focus on.

- **Formula** $VAC = BAC - EAC$
- **Positive VAC** Projected under-run. Green.
- **Negative VAC** Projected overrun. Red. Threshold typically $\pm 10\text{-}15\%$ of BAC triggers formal notification.
- **PM Action** Negative VAC requires a credible recovery plan or formal EAC rebaseline.

Indices and Completion Forecasts

Indices normalize performance to ratios. Forecasts project where you will end up.

CPI: Cost Performance Index $CPI = BCWP \div ACWP$

Cost efficiency: cents of value per dollar spent

CPI is the most important predictive metric in EVM. Once it drops below 1.0 after 20% completion, it almost never recovers.

- **CPI = 1.0** On budget: earning exactly what you are spending
- **CPI = 0.85** For every \$1 spent, only \$0.85 of value earned. 15% overrun rate.
- **CPI = 1.15** For every \$1 spent, \$1.15 of value earned. 15% under-run.
- **Humphreys Rule** Research shows cumulative CPI below 1.0 after 20% completion virtually never recovers. Early CPI trends are critical.

SPI: Schedule Performance Index $SPI = BCWP \div BCWS$

Schedule efficiency: how much work completed vs planned

SPI below 1.0 means falling behind plan. Loses meaning after 60% contract completion.

- **SPI = 1.0** On schedule.
- **SPI = 0.80** Only completing 80% of planned work.
- **Limitation** SPI approaches 1.0 at completion regardless of lateness. Use Earned Schedule (ES) for programs past 60% complete.
- **Use With** Cross-reference SPI with the network schedule critical path for a complete picture.

EAC: Estimate at Completion **MOST SCRUTINIZED NUMBER**

What the total program will actually cost

EAC is the contractor best estimate of total cost to complete all authorized work. Compare contractor EAC to statistical EAC (BAC/CPI); large gaps are red flags.

- **EAC = BAC / CPI** Statistical formula. Most accurate after 20% completion. Assumes future efficiency equals cumulative CPI.
- **EAC = ACWP + ETC** Bottom-up. Most accurate when fundamental change occurred. Most labor-intensive.
- **EAC = ACWP + (BAC – BCWP)** Assumes all overruns to date are non-recurring. Only valid for confirmed one-time events.
- **Government PM Rule** Always track BOTH contractor EAC AND BAC/CPI statistical EAC. Require explanation when they diverge more than 5%.

ETC: Estimate to Complete $ETC = EAC - ACWP$

How much MORE to finish remaining work

ETC is the projected cost for all work remaining. $EAC = ACWP + ETC$.

- **Formula** $ETC = EAC - ACWP$
- **Bottom-Up ETC** Control Account Managers (CAMs) estimate remaining work packages. Required after significant re-planning.
- **Sanity Check** Compare ETC to remaining authorized budget (BAC – BCWP). If $ETC \gg$ remaining budget, a formal overrun notification is likely required.
- **Watch For** ETC identical to remaining budget often means the estimate has not been updated: not that the program is healthy.

TCPI: To-Complete Performance Index $TCPI = (BAC - EV) \div (BAC - AC)$

Efficiency needed on all remaining work

TCPI > 1.10 means remaining work must be performed 10%+ more efficiently than historical average: usually unrealistic.

- **Formula vs BAC** $TCPI(BAC) = (BAC - BCWP) / (BAC - ACWP)$
- **Formula vs EAC** $TCPI(EAC) = (BAC - BCWP) / (EAC - ACWP)$
- **TCPI > 1.10** Requires performing 10%+ more efficiently than history. Likely unrealistic: EAC may be understated.
- **TCPI ≤ CPI** Recovery is mathematically feasible. Still needs a credible corrective action plan.

Everything else in EVM derives from these six. Know them cold.

INTERMEDIATE**Using EVM Indices Together: The 60-40 Rule and Beyond**

Mid-career PMs know that no single EVM index tells the full story. The most robust EAC formula weights CPI and SPI: $EAC = BAC \div (0.2 \times SPI + 0.8 \times CPI)$. This formula reflects the empirical finding that cost performance has more predictive power than schedule performance for final cost outcomes. But use indices in combination: a program with CPI 0.95 and SPI 0.85 has both a cost problem and a schedule problem. The schedule pressure will likely drive overtime and accelerated work that further degrades CPI. Programs with $CPI \geq 1.00$ but $SPI < 0.85$ are common: they appear cost-efficient but the schedule pressure is a leading indicator of future cost overruns. Don't let a healthy CPI lull you into ignoring a deteriorating SPI.

One Program, Walked All the Way Through

Acronyms stick once you have pushed a single program through every formula. Take a \$20M contract, twelve months, with work planned evenly at about \$1.67M a month. At the end of month six the plan said \$10M of work should be done (BCWS). The contractor has actually completed \$8M worth of the planned work (BCWP). Getting that far has cost \$9M (ACWP). Everything else is arithmetic on those three numbers, and the table below does all of it. Work through it once with a pen and the acronyms stop being a list to memorize.

Every Metric From Three Numbers (BAC \$20M, month 6)

METRIC	FORMULA	THIS PROGRAM	WHAT IT SAYS
Cost Variance	BCWP - ACWP	\$8M - \$9M = -\$1M	A million dollars over budget for the work done
Schedule Variance	BCWP - BCWS	\$8M - \$10M = -\$2M	Two million dollars of planned work not done
CPI	BCWP / ACWP	8 / 9 = 0.89	89 cents of work per dollar spent
SPI	BCWP / BCWS	8 / 10 = 0.80	Running at 80 percent of planned pace
EAC (statistical)	BAC / CPI	20 / 0.89 = \$22.5M	Where it lands if efficiency holds
ETC	EAC - ACWP	22.5 - 9 = \$13.5M	Cost still to come
VAC	BAC - EAC	20 - 22.5 = -\$2.5M	Projected overrun
TCPI (to BAC)	(BAC-BCWP) / (BAC-ACWP)	12 / 11 = 1.09	Must run at 1.09 to still hit \$20M

Three Ways to Compute EAC, and When Each One Is Honest

The statistical EAC above assumes the future looks like the past, which research has repeatedly found to be the safest default once a contract is roughly 20 percent complete. The second method is the contractor's bottom-up EAC, rebuilt from the remaining work package by package. It is more informed and reliably more optimistic, because the people estimating are the same people who will be judged on it. The third weights both: BAC divided by the product of CPI and SPI, used when schedule pressure is driving cost, since recovering a late schedule usually costs money. On our program those give \$22.5M statistical, whatever the contractor's build-up says, and \$28.1M on the CPI times SPI method. Present the spread, not one number, and say which assumption each rests on.

Where These Numbers Live in the Report You Receive

The formats of the Integrated Program Management Report carry these values in specific places, and knowing which format answers which question saves an hour every month. Format 1 gives the work breakdown structure view: BCWS, BCWP, ACWP and the variances by element, which is where you find what is driving the problem. Format 2 gives the same by the contractor's organization, which tells you which functional group owns it. Format 3 is the baseline and its changes, which is where rebaselining hides. Format 4 is staffing, the best leading indicator in the whole package. Format 5 is the narrative, where the contractor explains variances and states corrective action. The next lesson works through reading them; the point here is that every acronym you just computed arrives on a specific page.

ADVANCED**THE CUMULATIVE CPI STABILITY FINDING AND WHAT IT MEANS FOR PMS**

Research by Christle, Bhatt and others (the "Christle Findings") established that once a program has expended 20% of its budget, the cumulative CPI rarely improves by more than 10%. This has profound implications: a program at 20% complete with a cumulative CPI of 0.85 will, with very high probability, complete at a CPI no better than 0.85. Meaning a 15%+ cost overrun is essentially locked in. Senior PMs use this finding to drive two actions: (1) early IBR. Challenge cost realism before 20% is spent; (2) early OTB/EAC revision. If CPI is deteriorating before the 20% mark, don't wait for it to stabilize. The longer you wait to acknowledge an overrun, the more the sunk cost fallacy traps the program. The cumulative CPI stability finding is the empirical basis for "fix it early or pay double."

Why SV Lies at the End, and What Earned Schedule Fixes

Schedule Variance is measured in dollars, which is the original sin of EVM schedule analysis. At completion the contractor has earned the entire budget and the entire budget was planned, so BCWP equals BCWS equals BAC and SV equals zero, no matter how late the program finished. A program can deliver a year behind and show perfect schedule performance on the last report. Earned Schedule solves this by converting earned value into time: find the point on the planned curve where the cumulative BCWS equals the BCWP you have actually earned, and the distance between that point and today, in months, is your schedule variance in time. On our example program, \$8M of earned value corresponds to the four-point-eight-month point on the plan, so at month six the program is roughly 1.2 months behind. That number keeps working right up to delivery, which is exactly when the dollar version stops.

Rebaselining, Over Target Baseline, and Reading the Reset

When a baseline no longer represents any achievable plan, measuring against it produces variances nobody learns anything from. The formal remedy is an Over Target Baseline: the government agrees the contractor may replan to a budget above the contract value, so future performance is measured against something real. An Over Target Schedule does the same for dates. Both require government approval and both reset the variances to zero, which is why they attract attention. The legitimate case is a program where the original plan was overtaken by events everyone acknowledges. The illegitimate case is a program using a reset to erase a performance record before a review. What a senior PM watches is the pattern: how many resets, how close to a milestone decision, and whether the new baseline's assumptions are any more defensible than the ones that just failed.

How EVM Gets Gamed, and the Cross-Checks That Catch It

Any number a contractor is evaluated on will eventually be managed, and EVM has a standard repertoire. Front-loading puts generous budget in early work packages so early CPI looks strong and the pain arrives

late. Retroactive replanning moves budget into periods where the work is already done. Level-of-effort work packages earn value simply by time passing, so converting discrete work to level of effort makes performance problems invisible. Schedule logic changes remove negative float instead of recovering it. The cross-checks are all comparisons rather than single metrics: CPI against cost per physical unit delivered, SPI against critical path float, the percentage of budget in level-of-effort packages over time, and the count of baseline changes per quarter. None of these is proof of anything. All of them are questions worth asking out loud in a review, and asking them early changes behavior more than finding them later ever does.

WHAT SENIOR PMS DO DIFFERENTLY

They compute EAC three ways and present the spread with its assumptions rather than a single number. They switch to Earned Schedule for schedule analysis in the last third of a program, when the dollar version stops working. They treat every baseline reset as a question about the new assumptions, not just a procedural step. And they cross-check every index against a physical measure, because a number under evaluation is a number under management.

One sentence to teach your team: **every EVM metric comes from three numbers, planned work, earned work, and actual cost, so once you can walk one program through the arithmetic the acronyms stop being vocabulary and start being a diagnosis.**

BUILD ON THIS

- **Reading IPMR Reports** Where each of these numbers arrives, by format
- **Estimate at Completion (EAC)** The forecast argument in depth
- **Revenue Recognition II** Why your EAC is also the contractor's revenue number

Key Terms

BAC

Budget at Completion. The total authorized budget for all work on the contract.

BCWS / PV

Budgeted Cost of Work Scheduled (Planned Value). What you planned to spend by today.

BCWP / EV

Budgeted Cost of Work Performed (Earned Value). The budget value of work actually completed.

ACWP / AC

Actual Cost of Work Performed. Real dollars spent on completed work, from the accounting system.

CPI

Cost Performance Index = $BCWP / ACWP$. Below 1.0 means over budget.

SPI

Schedule Performance Index = $BCWP / BCWS$. Below 1.0 means behind schedule.

EAC

Estimate at Completion. Projected total cost to finish all authorized work.

ETC

Estimate to Complete. Projected cost to finish remaining work only.

VAC

Variance at Completion = $BAC - EAC$. Negative means projected overrun.

TCPI

To-Complete Performance Index. Efficiency needed on remaining work to hit BAC or EAC.

Knowledge Check

A program has BCWP = \$800K, ACWP = \$1,000K, BCWS = \$900K. What is the CPI?

A. 0.89

✓ B. 0.80

C. 1.25

D. 1.13

WHY

$CPI = BCWP / ACWP = \$800K / \$1,000K = 0.80$. For every dollar spent, only \$0.80 of value is earned. A 20% cost overrun rate.

Using the same data (BCWP=\$800K, ACWP=\$1,000K, BCWS=\$900K), what is the Schedule Variance?

A. +\$100K (ahead)

✓ B. -\$100K (behind)

C. -\$200K (behind)

D. +\$200K (ahead)

WHY

$SV = BCWP - BCWS = \$800K - \$900K = -\$100K$. Negative SV means behind schedule, \$100K worth of planned work has not been completed.

What does TCPI > 1.10 indicate?

A. Program is 10% ahead of budget

✓ B. Remaining work must be performed 10%+ more efficiently than history: likely unrealistic

C. Program has a 10% cost overrun

D. Program needs only 90% of remaining budget

WHY

TCPI of 1.10 means to stay within BAC, future work must be performed 10% more efficiently than the cumulative CPI. When TCPI significantly exceeds current CPI, the EAC should be revised upward.

The statistical EAC formula (BAC/CPI) is most reliable when:

- A. Contract is in its first month
- ✓ B. After approximately 20% contract completion
- C. Only when contractor provides bottom-up estimate
- D. During final 10% of contract

WHY

Research (RAND, GAO) consistently shows EAC = BAC/CPI is among the most accurate predictors after approximately 20% completion. Often more accurate than optimistic bottom-up estimates.

Why does Schedule Variance (SV) converge to zero at contract completion?

- A. All overruns are corrected at completion
- ✓ B. BCWP and BCWS both equal BAC at completion, making SV = 0 regardless of lateness
- C. Contractor reconciles variances before final payment
- D. Government accepts any date within 10% of plan

WHY

At contract completion all work is earned (BCWP = BAC) and all was planned (BCWS = BAC), so SV = 0 regardless of how late the program finished. Use Earned Schedule (ES) for late-program schedule analysis.

On the worked program (BAC \$20M, BCWS \$10M, BCWP \$8M, ACWP \$9M), what is the ETC using the statistical EAC?

- A. \$11.0M
- B. \$12.0M
- ✓ C. \$13.5M
- D. \$20.0M

WHY

EAC is $BAC/CPI = 20/0.89 = \$22.5M$. ETC is EAC minus ACWP: $\$22.5M - \$9M = \$13.5M$.

Why is the contractor's bottom-up EAC usually more optimistic than the statistical one?

A. It uses different accounting rules

✓ B. The people rebuilding the estimate are the same people who will be judged on it

C. It excludes indirect costs

D. It is calculated less often

WHY

It is better informed and structurally optimistic. Present it alongside the statistical figure rather than instead of it.

Which IPMR format is the best leading indicator of future performance?

A. Format 1, the work breakdown view

B. Format 3, the baseline

✓ C. Format 4, staffing

D. Format 5, the narrative

WHY

Staffing moves before cost and schedule do. A ramp that does not materialize shows up in Format 4 months before it shows up in CPI.

A program delivers a year late. What does its final Schedule Variance show?

A. A large negative number

✓ B. Zero

C. The number of months late

D. It cannot be calculated

WHY

At completion BCWP and BCWS both equal BAC, so SV is zero regardless of lateness. This is why Earned Schedule exists.

What does an Over Target Baseline do?

A. Reduces the contract value

✓ **B. Lets the contractor replan to a budget above contract value, with government approval, so variances measure something achievable**

C. Cancels the EVM requirement

D. Transfers cost risk to the government automatically

WHY

It resets variances to zero, which is why the pattern of resets and the credibility of the new assumptions matter more than the mechanics.

Order the calculation from the three source numbers to a projected overrun:**CORRECT ORDER**

1. Collect BCWS, BCWP and ACWP for the period
2. Compute the variances: $CV = BCWP - ACWP$ and $SV = BCWP - BCWS$
3. Compute the indices: $CPI = BCWP / ACWP$ and $SPI = BCWP / BCWS$
4. Project EAC, most defensibly as BAC / CPI
5. State $VAC = BAC - EAC$ as the projected overrun or underrun

WHY

Variances give magnitude, indices normalize it, EAC projects it, and VAC states the overrun.

Match each gaming pattern to the cross-check that catches it:**CORRECT MATCHES**

Front-loaded budget in early work packages	CPI against cost per physical unit delivered
Schedule logic changed to remove negative float	SPI against critical path float trend
Discrete work converted to level of effort	Share of budget in level-of-effort packages over time
Retroactive replanning	Count of baseline changes per quarter

WHY

Each check pairs a reported number with a physical or structural one, which is the only way these show up early.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 4

Reading IPMR Reports

22 min

Learn to read and analyze the six IPMR formats that contractors submit monthly on DoD programs, and understand what every government PM should look for in each one.

What Is the IPMR and Why Does It Matter?

The Integrated Program Management Report (IPMR) is the primary contractually required data deliverable for EVM reporting on DoD contracts. It replaced the older Contract Performance Report (CPR) and Contract Funds Status Report (CFSR) through DI-MGMT-81861 (2012). If your contract exceeds \$50M cost or incentive type (raised from \$20M in Feb 2026 under the DFARS overhaul. Validated EVMS is now required above \$100M, raised from \$50M), you will receive IPMR data from your contractor every month. This lesson teaches what each IPMR format contains and what government PMs look for.

The Six IPMR Formats

Format 1: WBS Performance Summary MOST USED

Cost and schedule performance rolled up by Work Breakdown Structure

Format 1 shows cumulative and at-completion cost and schedule performance organized by WBS element. This is the first format PMs open: it shows where the money and schedule problems are.

What to Look for in Format 1

- **WBS elements with negative CV (Cost Variance): over-budget areas. Drill down to understand root cause.** Correct. The most actionable Format 1 analysis starts with CV: negative CV means $AC > EV$, indicating a cost overrun. Sorting by CV magnitude identifies where program dollars are most at risk. Drill down to CAP (Control Account Plan) level to identify root causes: scope growth, rework, inefficiency, or inaccurate estimates.
- **WBS elements with negative SV (Schedule Variance): behind-schedule areas. Cross-check against the IMS.** Correct. Negative SV ($EV < PV$) indicates the program is behind schedule in EV terms. Cross-reference Format 1 SV against the IMS to identify whether schedule slippage is on the critical path or in float-absorbing near-critical tasks.
- **BAC vs. EAC at the total contract level: the delta is VAC. A growing negative VAC trend is the most critical warning sign.** Correct. VAC (Variance at Completion) = $BAC - EAC$. A growing negative VAC trend: month over month: is the single most important indicator of program cost health. A program with declining CPI that does not adjust its EAC is masking the true projected overrun.
- **MR (Management Reserve) balance: if MR is eroding rapidly, the contractor is using contingency to mask overruns.** Correct. MR is budget held outside the PMB for unplanned in-scope work. Rapid MR burn-down: especially without corresponding PMB growth or new work authorization: may indicate the contractor is using MR to absorb overruns instead of properly recognizing negative variance.
- **Undistributed Budget (UB): budget not yet assigned to WBS elements. Large UB balances late in the program are a concern.** Correct. UB represents authorized but unplanned work: budget not yet assigned to specific Control Accounts. Large UB balances in the middle or late phases of a program suggest poor planning, inadequate work authorization, or potential use of UB as a hidden reserve.

Format 1 Red Flags

- {'label': 'CPI < 0.90', 'value': 'At-completion overrun of over 10% is likely. Initiate Over Target Baseline (OTB) discussion if CPI is stable at this level.'}
- {'label': 'EAC < Statistical EAC', 'value': 'Contractor may be sandbagging risk. Compare EAC vs BAC/CPI and require explanation for gaps over 5%.'}
- {'label': 'Rapidly Decreasing MR', 'value': 'Management Reserve burning faster than planned indicates undocumented risk realization.'}
- {'label': 'Large UB Late in Program', 'value': 'Undistributed budget that should have been assigned months ago may indicate poor planning or intentional delay.'}

Format 2: OBS Performance Summary **ORGANIZATIONAL VIEW**

Performance by organizational element and subcontractor

Format 2 shows performance by responsible organization (divisions, departments, subcontractors). Use it to identify which teams or subcontractors are driving cost and schedule problems.

What to Look for in Format 2

- **Which organizational element has the worst CPI: often points to staffing or technical problems in that org** Correct. Format 2 breaks performance data by organizational reporting elements (OREs): the contractor's internal organizational structure. The ORE with the worst CPI is typically experiencing a localized problem: understaffing, technical difficulty, management issues, or subcontractor problems within that organizational unit.
- **Subcontractor performance: if the prime's work is on track but subcontractors are in trouble, supply chain management needs scrutiny** Correct. Prime contractor CPR/IPMR data may mask subcontractor-level problems. If specific WBS elements (often subcontractor-heavy) show consistent negative variance, the prime's supply chain management requires scrutiny. Government may request subcontractor-level EVM data for significant subcontracts.
- **Compare Format 2 performance by org against staffing data: declining BCWP with high ACWP from a specific org often means unproductive labor** Correct. When an organization shows declining BCWP (accomplishment) but sustained or increasing ACWP (spending), it indicates labor investment is not translating to earned work: a classic inefficiency or management problem indicator.

Format 3: Baseline BASELINE HEALTH

Time-phased PMB and how it has changed: baseline health

Format 3 shows the performance measurement baseline over time: including changes, management reserve draws, and budget reallocations. This is how you detect baseline instability.

What to Look for in Format 3

- **Baseline changes from last period: any retroactive replanning (moving budget backward to cover past variances) is a serious concern** Correct. Retroactive replanning: changing the PMB to cover past variances: is prohibited without explicit government approval. Format 3 changes must always be prospective; government approval is required for significant baseline changes.
- **Management Reserve (MR) draw-downs: compare total MR used against original MR balance** Correct. Calculate the MR burn rate: $(\text{Original MR} - \text{Current MR Balance}) / \text{Months elapsed} = \text{Average MR burn per month}$. At the current rate, how many months until MR is exhausted? Premature MR exhaustion forces the PM into difficult tradeoffs.
- **Budget shifts between near-term and far-term, "rubber baseline" behavior is a gaming indicator** Correct. 'Rubber baseline' behavior: repeatedly moving planned value (BCWS/PV) further into the future to avoid recording negative schedule variance: is a form of baseline gaming. The baseline 'stretches' to accommodate actual performance instead of recording the true schedule delay.
- **Current PMB vs. original PMB: large deviations indicate significant re-planning requiring government approval** Correct. The PMB should only change through formally approved processes: Engineering Change Proposals (ECPs), Over-Target Baseline (OTB) approval, or government-approved replanning. Large deviations from the original PMB without formal authorization indicate improper baseline management.

Retroactive changes to the PMB require Government approval. Any change reaching back more than 1-2 reporting periods is suspect. Over Target Baseline (OTB) reprogramming is the formal mechanism for resetting an unrealistic baseline. It requires government approval and resets the measurement zero.

Format 4: Staffing WORKFORCE

Actual vs. planned headcount by labor category: a leading indicator

Format 4 shows planned and actual staffing by labor category (LCATs). It is often the leading indicator of future cost and schedule problems.

What to Look for in Format 4

- **Understaffing vs. plan: if actual headcount is significantly below planned, BCWP will lag BCWS (negative SV). Schedule slippage is predictable.** Correct. Format 4 provides staffing data by labor category. Comparing actual headcount to planned identifies the root cause of SV in Format 1. Understaffing is the most common and predictable schedule driver: if the contractor is not putting the people on the program, the work cannot get done on schedule.
- **Overstaffing vs. plan: more labor than planned drives ACWP above BCWP (negative CV). May indicate rework or underestimated complexity.** Correct. Overstaffing: more actual labor than planned: drives ACWP up without corresponding BCWP gain if the additional labor is not producing incremental value. Common causes: rework from failed tests, underestimated task complexity, or inefficient work authorization.
- **LCAT mix changes: substituting lower-grade labor for planned senior roles may affect technical quality** Correct. Labor category mix shifts can affect both cost and technical quality. Substituting junior staff for planned senior engineers may reduce ACWP short-term but increase rework costs and schedule risk if junior staff cannot perform at the required technical level.
- **Cross-reference Format 4 with Format 1: understaffed WBS elements will show negative SV in Format 1** Correct. This cross-format analysis is one of the most powerful tools in CPR review. If Format 4 shows understaffing in a specific organizational element, and Format 1 shows negative SV in the corresponding WBS element, the causal relationship is confirmed.

Format 5: Explanations and Problem Analysis MOST IMPORTANT

Variance narratives: the story behind the numbers

Format 5 contains the contractor's written analysis of significant cost and schedule variances. A well-written

Format 5 explains root cause, impact, and corrective action.

What to Look for in Format 5

- **Causality, not just description, "labor hours exceeded plan due to test failures" is better than "costs increased"** Correct. The most common Format 5 weakness is descriptive narratives that describe what happened without explaining why. Government reviewers expect causal analysis: what specific event or condition caused the variance? Root cause quality is the primary criterion for Format 5 adequacy.
- **Corrective action specificity: vague actions ("team is working the issue") with no timeline are not acceptable** Correct. Acceptable corrective actions name a specific action, identify an owner by name and position, and include a target completion date. 'The deputy PM is implementing a revised subcontractor surveillance plan, with monthly performance reviews beginning March 15' is acceptable. 'We are working the issue' is not.
- **Recovery schedule: if behind schedule, Format 5 should show how the contractor plans to recover** Correct. For WBS elements with negative SV, Format 5 should include a recovery plan: specific actions to recover the schedule slip. If recovery is not achievable, the narrative should acknowledge that and quantify the expected schedule impact.
- **EAC rationale: the Format 5 should explain the basis for the contractor's EAC, especially if it differs from statistical EAC** Correct. The contractor's EAC must be explained in Format 5 when it differs materially from statistical EAC methods (BAC/CPI). If the contractor's EAC is more optimistic than statistical methods predict, the government should require a detailed justification.
- **Variance thresholds: ensure all threshold variances (typically CV or SV over plus or minus 10% and over \$100K) have narratives** Correct. Most government contracts specify variance thresholds (e.g., CV or SV >10% and >\$100K) that require written narratives in Format 5. Inconsistency in threshold reporting is itself a compliance finding.

Format 5 Scrutiny Checklist: (1) Is root cause specific and credible? (2) Are corrective actions assigned, dated, and tracked? (3) Does the EAC reflect the corrective action cost? (4) Are past period corrective actions from last month's F5 showing results in this month's F1? (5) Does the narrative acknowledge schedule impact, not just cost impact?

Format 6: IMS / Schedule SCHEDULE

The Integrated Master Schedule in electronic format

Format 6 is the contractor's Integrated Master Schedule (IMS) as an electronic file (Primavera P6 or MS Project). It shows network logic, critical path, and near-term milestones.

What to Look for in Format 6

- **Critical path: any slip on critical path tasks directly delays contract completion** Correct. The integrated master schedule critical path is the sequence of tasks with zero total float. Any delay to a critical path task delays the program end date by the same amount. Government reviewers focus on critical path health, stability, and the float status of near-critical tasks.
- **Total Float: tasks with zero or negative total float are on or near critical path. Negative float means the task is already late.** Correct. Total float represents how much a task can be delayed without delaying the program end date. Zero total float = critical path. Negative total float = the task's schedule is already late relative to the plan. Negative float is a direct indicator of embedded schedule problems.
- **Schedule density: unusually high milestone clusters ("schedule packing") are a sign of unrealistic planning** Correct. Schedule packing: an unrealistic concentration of milestones or completion events in a specific period; suggests the schedule was built backward from a desired end date rather than built bottom-up from realistic work estimates. It is a planning integrity red flag.
- **Logic ties: all tasks should have predecessor/successor relationships. Tasks with no ties cannot drive accurate critical path analysis.** Correct. Every task in a well-structured IMS should have at least one predecessor and one successor. Tasks with no predecessor/successor relationships: called 'open-ended tasks' or 'dangling activities': float freely and do not contribute to a realistic critical path analysis.
- **Compare current IMS critical path to last period: an unstable critical path indicates poor schedule management** Correct. An unstable critical path: one that shifts significantly month to month: indicates poor schedule management or deliberate baseline gaming. Major shifts suggest the schedule is being manipulated rather than updated based on actual progress.

IPMR Monthly Review Workflow: 6-Step Process (~50 min)

Step 1: Check Overall Contract Health 5 MIN

Format 1: Total contract CPI and VAC trend

Start at the top: is the program over or under budget overall? Which direction is the trend moving?

What to look for

- **CPI < 1.0: cost overrun: how far, how fast is it moving?** Correct. $CPI = EV/AC$. A CPI below 1.0 means the program is spending more than the budgeted value for work accomplished. The critical questions: (1) How far below 1.0? (0.95 is concerning; 0.80 is severe). (2) Is the trend worsening month over month? (3) Has it crossed the 20% completion point?
- **VAC (negative): projected overrun at completion: is it growing each month?** Correct. $VAC = BAC - EAC$. A growing negative VAC trend: month over month: is the most critical program cost health indicator. If VAC deteriorates by \$5M each month report, the program office needs to act before the overrun becomes unmanageable.
- **SPI < 1.0: behind schedule: note which months it started declining** Correct. $SPI = EV/PV$. $SPI < 1.0$ indicates schedule slippage in EV terms. Identifying when SPI started declining is critical: was it a specific event (test failure, key personnel departure) or a slow degradation?
- **Red flag: CPI trending down 3+ months in a row = systemic problem, not a blip** Correct. Three consecutive months of declining CPI is the strongest indicator of a systemic program problem. Research by DAU and Paul Solomon shows that after three months of declining CPI, the trend almost never reverses significantly. At this point, $EAC = BAC/CPI$ is the most reliable predictor.

Step 2: Identify the Worst WBS Elements 10 MIN

Format 1: Top 3 elements by worst Cost Variance (CV) and Schedule Variance (SV)

Drill past the summary: find the specific work packages bleeding cost and schedule.

What to look for

- **Sort CV column: the largest negative CV numbers are your biggest overruns in dollars** Correct. Sorting by CV magnitude identifies the WBS elements consuming the most excess cost. Cross-reference against element size (BAC), a -\$5M CV on a \$10M element (50% overrun) is more alarming than a -\$5M CV on a \$100M element.
- **Sort SV column: the largest negative SV numbers are your most schedule-critical slips** Correct. After CV, sort SV to identify which WBS elements are furthest behind schedule. Cross-reference against the IMS to determine whether these elements are on the critical path.
- **Note if the same WBS element appears in both CV and SV worst lists: that is a compounding problem** Correct. A WBS element appearing in both the worst CV and worst SV lists is a compounding problem: spending more than planned AND achieving less than planned simultaneously. These dual-variance elements pose the highest program risk and require immediate management attention.
- **Compare to last month: are the same elements getting worse, or are new ones appearing?** Correct. Trend analysis is more informative than single-period snapshots. The same elements getting progressively worse indicates an uncorrected systemic problem. New elements appearing suggests scope growth or new technical challenges, a broader problem.
- **Cross-check against Format 5: do these elements have variance narratives? If not, ask why.** Correct. Every WBS element exceeding variance thresholds must have a Format 5 narrative. If high-variance elements lack narratives, it indicates either inadequate EVM reporting compliance or the contractor does not understand the significance of the variance.

Step 3: Read the Variance Narratives 15 MIN

Format 5: Narratives for all threshold variances: the most important 15 minutes

Numbers tell you what happened. Narratives tell you why; and what the contractor plans to do about it.

Quality checks for every narrative

- **Does the root cause actually explain the CV/SV? Vague causes ("resource constraints") are a red flag.** Correct. 'Resource constraints' is the most commonly used vague root cause in Format 5, it explains nothing specific. What resources? What constraint? Why was it not anticipated? Government reviewers should push back on vague narratives.
- **Is there a specific, measurable corrective action with a named owner and due date?** Correct. Corrective actions must be SMART: Specific, Measurable, Assigned (who is responsible), Realistic, and Time-bound (by when). Corrective actions without owners and due dates are aspirational, not commitments.
- **Does the corrective action realistically address the root cause?** Correct. A common trap is well-written corrective actions that do not address the identified root cause. If the root cause is 'test failures due to software integration problems,' a corrective action of 'increase staffing' may not solve the underlying software issue.
- **Mismatch check: does the narrative address the same WBS elements with the biggest variances in Format 1? Inconsistency = narratives written independently of the data.** Correct. A frequent quality issue: Format 5 narratives addressing WBS elements with minor variance while high-variance elements go unexplained. This indicates the narratives were written independently without cross-referencing Format 1 data.
- **Track last month's corrective actions: did they produce improvement? If CPI did not move, the action was ineffective.** Correct. Corrective action effectiveness tracking is the most important longitudinal review skill. If the PM committed to a specific corrective action last month, did CPI/SPI improve for that WBS element this month? Repeated commitments with no measurable improvement indicate the actions are ineffective.

Step 4: Check for Baseline Changes 5 MIN

Format 3: Baseline Log: any unauthorized re-baselining?

Contractors sometimes hide cost overruns by re-baselining work packages. Catch it here.

What to look for

- **Any new entries since last month? All changes require CO approval via formal contract modification.** Correct. Format 3 documents all changes to the PMB. Any significant change requires government approval. Reviewing Format 3 for unauthorized changes, new entries without corresponding contract modifications, is an important compliance check.
- **Budget transfers between WBS elements: legitimate re-planning or hiding overruns in other accounts?** Correct. Internal replanning can be used legitimately to reflect refined work planning. However, it can also be used to spread overruns across multiple elements to keep individual elements below reporting thresholds. Large unexplained budget transfers should trigger questions.
- **Management Reserve (MR) draws: how much MR remains? MR burn rate is a key program health indicator.** Correct. Calculate the MR burn rate: $(\text{Original MR} - \text{Current MR Balance}) / \text{Months elapsed} = \text{Average MR burn per month}$. If MR will be exhausted significantly before contract completion, the program faces difficult choices.
- **Undistributed Budget (UB): large UB that's been sitting means work is not yet planned in detail: schedule risk.** Correct. UB represents authorized work not yet planned at the Control Account level. Large, persistent UB late in a program indicates work is not being planned in detail: which means schedule cannot be tracked, and the PMB understates the full scope of planned activity.

Step 5: Validate Staffing vs. Recovery Plans 5 MIN

Format 4: Staffing: does headcount actually support the corrective actions in Format 5?

Contractors often commit to recovery plans that require staff they do not have. This is where you catch it.

What to look for

- **Cross-reference: which WBS elements have negative SV in Format 1? Are those same elements fully staffed in Format 4?** Correct. This cross-format analysis confirms staffing causality. If Format 1 shows negative SV on Element X, and Format 4 shows Element X is staffed at 60% of plan, the causal relationship is clear.
- **Understaffed + negative SV = schedule slip will continue regardless of recovery commitments** Correct. If a WBS element is significantly understaffed AND showing negative SV, no amount of 'working the issue' will recover the schedule until staffing reaches plan levels. Recovery is mathematically impossible without additional resources.
- **Check key personnel: are the named experts from the proposal actually on the program?** Correct. Key personnel commitments made in the winning proposal are often contractually binding. If the proposal named a specific PM or Deputy PM, and that person is not on the program, the contractor may be in breach of contract.
- **Headcount trending down while schedule is behind = problem getting worse, not better** Correct. A declining headcount trend combined with negative SV is a strongly negative leading indicator. It means the contractor is not just behind, they are further reducing the resources needed to recover.

Step 6: Check the Critical Path 10 MIN

Format 6: IMS: compare critical path to the milestone plan

If the critical path slips, the program end date slips. No exceptions.

What to look for

- **Has the critical path changed since last month? New tasks appearing on the critical path = scope risk emerging** Correct. Critical path stability is a key schedule health indicator. New tasks appearing on the critical path indicate either previously float-rich tasks have consumed their float, or the contractor is manipulating the schedule.
- **Total float on near-critical tasks: anything under 10 days of float is effectively critical** Correct. Tasks with very low total float (10 days or fewer) are operationally critical, any minor disruption will push them onto the critical path. Monitoring near-critical float consumption is an early warning system for emerging critical path issues.
- **Are key milestones (CDR, PDR, IOT&E) still tracking to the APB schedule?** Correct. Key program milestones, CDR, PDR, IOT&E, are anchored in the APB and are the primary schedule commitments to oversight stakeholders. Slipping these milestones triggers APB reporting and potential Nunn-McCurdy concerns.
- **Negative total float (tasks already late): how many, and are they on the path to a contract milestone?** Correct. Negative total float is an unambiguous indicator of existing schedule problems embedded in the baseline. If negatively-floated tasks feed into a contract milestone, that milestone is already late in practice, whether anyone has flagged it or not.
- **Compare to Format 5: are schedule slips on the critical path explained in the variance narratives?** Correct. Critical path schedule slippage must be addressed in Format 5 narratives. If the IMS shows critical path tasks with negative SV and Format 5 narratives for those WBS elements are missing or vague, this is a significant compliance gap.

INTERMEDIATE**Cross-Referencing IPMR Formats: The Monthly Analysis Workflow**

Mid-career PMs don't just read IPMR formats sequentially. They cross-reference them. The highest-value cross-references: (1) Format 1 variance WBS elements → Format 5 narrative: does the narrative actually address the Format 1 problem areas, or does it address different elements? Inconsistency means the Format 5 was written independently of the data; (2) Format 4 staffing gaps → Format 1 schedule variance: understaffed WBS elements should show negative SV. If they don't, the schedule logic is disconnected from staffing reality; (3) Format 5 corrective action commitments → next month's Format 1: are last month's corrective actions producing measurable improvement? If not, the corrective action was ineffective or was not actually executed. Build a monthly tracking sheet that captures the key cross-references from each IPMR submission.

ADVANCED

IPMR AS LEGAL DOCUMENTATION: WHAT SENIOR PMS MUST PRESERVE

Senior PMs on large programs must understand that IPMR data is contractually required documentation and is discoverable in litigation. Format 5 narratives become exhibits in ASBCA cases. A contractor's written explanation that a variance is due to "government-caused delays" is the foundation of a future REA. The government PM's obligation: (1) preserve all IPMR submissions with receipt timestamps; (2) document government responses in writing. If you accepted an inadequate Format 5 narrative without objection, you may have waived your right to challenge the root cause later; (3) track OTB and reprogramming history. Baseline changes require formal government approval and must be documented with the approving official's name and date. Programs that lose IPMR archives lose their ability to reconstruct performance history in claims disputes.

One sentence to teach your team: **the IPMR is six views of the same program, and the value is never in one format, it is in whether Format 1's variances, Format 5's narratives and Format 6's schedule tell the same story.**

BUILD ON THIS

- **Earned Value Management** The measurement system the IPMR reports
- **Estimate at Completion** Where the IPMR's forecast columns come from
- **Monthly Status Reports** The narrative report that must agree with the IPMR

Key Terms

IPMR

Integrated Program Management Report. The primary monthly EVM data deliverable from contractors on DoD programs, replacing the older CPR and CFSR.

Format 1

WBS Performance Summary. The most used IPMR format showing cost and schedule performance by WBS element.

Format 5

Explanations and Problem Analysis. Written narratives from the contractor explaining root causes of variances and corrective actions.

Format 6

Integrated Master Schedule (IMS) in electronic format. Shows network logic, critical path, and near-term milestones.

OTB

Over Target Baseline. A formal government-approved action that resets the Performance Measurement Baseline when the current baseline is no longer achievable.

Total Float

The amount of time a task can slip before it delays a downstream milestone or the contract completion. Negative total float means the task is already late.

VAC

Variance at Completion. The difference between BAC and EAC. Negative VAC means the program is projected to overrun budget.

Format 2

IPMR data organized by the contractor's organizational breakdown structure rather than by work. It shows which functional organization is driving a variance.

Performance Measurement Baseline (PMB)

The time-phased budget plan against which performance is measured. Changes to it are the first thing a reviewer checks.

Undistributed Budget

Authorized work and budget not yet allocated to control accounts. A large or long-lived balance here is a warning sign about planning maturity.

Knowledge Check

Which IPMR format contains written narratives explaining root causes and corrective actions for significant variances?

- A. Format 1: WBS Performance
- B. Format 3: Baseline

✓ C. Format 5: Explanations and Problem Analysis

- D. Format 6: IMS/Schedule

WHY

Format 5 contains the Variance Analysis Reports (VARs). Written narratives that explain the root cause of significant cost and schedule variances and describe corrective actions. It is the most important format for understanding WHY performance is off track.

A government PM reviewing Format 3 notices the PMB was retroactively changed to reduce a prior period's negative SV. This most likely indicates:

- A. Normal baseline maintenance per EVMS guidelines

✓ B. A potential EVMS compliance violation: retroactive replanning to conceal schedule variances

- C. An approved change order requiring replanning
- D. A subcontractor claiming additional scope

WHY

Retroactive changes to the PMB to reduce prior period variances are a serious EVMS compliance concern. ANSI 748 and DoDI 5000.02 prohibit retroactive baseline revisions except in very specific circumstances. This is classic "rubber baseline" gaming behavior.

Format 4 shows actual headcount is 30% below planned for the critical design phase. What should you expect in Format 1?

A. Positive CPI due to lower labor costs

✓ B. Negative SV because insufficient staff means less work is being accomplished

C. Positive SV because fewer people compresses the schedule

D. No impact: staffing and EVM performance are independent

WHY

Understaffing is a direct cause of negative Schedule Variance. Fewer people completing work means BCWP accumulates slower than BCWS, creating negative SV. Cross-referencing Format 4 staffing gaps with Format 1 schedule variances is a fundamental IPMR analysis technique.

In Format 6 (IMS), a task shows "-15 days total float." What does this mean?

A. The task has 15 days of scheduling flexibility

✓ B. The task is already 15 days late relative to its constraint and will delay the program end date

C. The task was completed 15 days early

D. The task needs 15 additional resources

WHY

Negative total float means the task is already past its planned completion date by that amount. Without recovery actions, negative float propagates through the network and delays all successor tasks including the contract completion milestone.

What is an Over Target Baseline (OTB) and when is it used?

A. An informal replanning action to reset performance metrics

✓ B. A formal government-approved reprogramming action that resets the PMB when the current baseline is no longer achievable, requiring a new IBR

C. A penalty mechanism for contractors who exceed the BAC

D. A supplemental funding authorization for overrun programs

WHY

An OTB is a formal, government-approved action that resets the Performance Measurement Baseline when the current baseline is so unrealistic that EVM measurements have lost informational value. It requires SSA/PEO approval, a revised EAC, and a new Integrated Baseline Review (IBR). OTB does not change the contract price.

Format 1 shows a \$4M unfavorable cost variance concentrated in one WBS element, but Format 5 explains it in general terms about 'labor challenges.' What is the reviewer's correct next step?

A. Accept the narrative

✓ B. Push for a root cause tied to that specific WBS element with a corrective action and a date, because a generic narrative on a concentrated variance means the cause has not been isolated

C. Request an Over Target Baseline

D. Reduce the fee

WHY

Format 5 exists to explain Format 1. A narrative that could apply to any program is a sign the analysis has not been done.

Why does a reviewer cross-reference Format 4 against Format 1 variances?

A. Format 4 shows profit

✓ B. Because staffing plans versus actuals often explain whether a cost variance is a rate problem, an efficiency problem, or simply work that has not been performed yet

C. Format 4 lists subcontractors

D. Format 4 sets the baseline

WHY

Headcount data separates 'we spent too much per hour' from 'we never staffed it,' and those two problems have completely different fixes.

A task in Format 6 shows negative total float. What does that mean?

A. The task has slack remaining

✓ B. The task is behind the point where it can still finish without delaying a constraint or the program finish date

C. The task is complete

D. The task is unfunded

WHY

Negative float is the schedule's way of saying the plan is already infeasible unless something is compressed.

What is the strongest reason a government PM should be cautious about approving an Over Target Baseline?

A. It costs money to prepare

✓ B. It resets the measurement baseline, so historical variance data loses comparability and the program's performance history effectively restarts

C. It requires a protest

D. It changes the contract type

WHY

An OTB can be the right call when the old baseline is meaningless, but it erases the trend line that would otherwise inform the forecast.

Which IPMR format would you consult first to determine whether a variance is concentrated in one functional organization rather than one work element?

A. Format 1

✓ B. Format 2

C. Format 5

D. Format 6

WHY

Format 1 is by work breakdown; Format 2 is by organization. Together they locate a problem in both dimensions.

A program's undistributed budget has stayed at 12 percent of contract value for eight months. What does an experienced reviewer conclude?

A. Excellent planning discipline

✓ B. Authorized work has not been planned into control accounts, so a meaningful share of the program is not actually being measured

C. The program is ahead of schedule

D. The contract needs an OTB

WHY

Undistributed budget is a parking lot. Work sitting there is not in any control account, which means nobody is earning or reporting value on it.

Why does Format 5 matter in a later dispute even though it carries no numbers?

A. It does not; only the numbers matter

✓ **B. Because the narratives are contractually required, contemporaneous documentation of cause and corrective action, and they are discoverable**

C. Because Format 5 sets the fee

D. Because Format 5 replaces the MSR

WHY

Contemporaneous written explanations carry unusual weight later, which is exactly why they should be written carefully the first time.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 5

Building a One-Page Program Dashboard

24 min

From raw program data to the single page leadership reads. What belongs on it and what does not, where each number comes from, how to build it in a spreadsheet or Power BI without a data team, the refresh discipline that keeps it trusted, and the design mistakes that make dashboards decorative instead of useful.

The Page Leadership Actually Reads

The Program Metrics lesson explained what to measure. This lesson is about presenting it: one page, refreshed on a schedule, that tells a reader with five minutes whether the program is healthy, where it is not, and what is being done. Most programs have several such pages, built by different people from different data, and they disagree. Leadership learns to trust none of them. The goal here is one page, from agreed sources, built by the program, that becomes the reference. It does not require a data team or an expensive tool. It requires deciding what belongs on it and keeping it honest.

What Belongs on a Program Dashboard

AREA	METRICS	SOURCE OF RECORD	REFRESH
Cost	Funded vs. spent vs. plan; burn rate vs. plan; EAC vs. budget; CPI if EVM applies	Accounting system; EVM system	Monthly, after close
Schedule	Milestones due, met, slipped (days); critical path status; SPI if EVM applies	The IMS	Weekly or biweekly
Performance	The two to four measures that define success: availability, throughput, defect rate, or the PWS standards	Operational systems; QASP surveillance	Per the measure; often weekly
Risk	Top five risks with rating and trend; new and closed this period	Risk register	Monthly, after the risk board
Staffing	Positions filled vs. required; clearances pending; key personnel changes	HR and program roster	Weekly
Deliverables	CDRLs due, submitted, accepted, overdue	Deliverable tracker	Weekly
Issues and decisions needed	Open issues with owner and date; decisions leadership must make	Action log	Weekly

What Does Not Belong

Everything else. A dashboard with forty metrics is a report, and nobody reads a report in five minutes. The test for each candidate metric: would a change in this number change a decision leadership would make? If not, it belongs in the appendix or the monthly status report, not on the page. Vanity metrics (number of meetings held, documents produced, hours worked) fail the test. Metrics the program cannot source reliably fail it too; a number that is estimated by hand every month is a number that will be wrong. And metrics without thresholds fail it, because a number with no definition of good or bad is decoration.

Building It: Five Steps That Work Without a Data Team

- **Decide the metrics and thresholds with the readers** Ask leadership what they need to decide and pick the metrics that inform those decisions. Set thresholds from the plan and the contract, in writing, before the first refresh.
- **Identify the source of record for each** One system per number. If two systems disagree, pick one and note it. If no system has the number, either build the process that produces it or drop the metric.
- **Build the data pull** A spreadsheet tab per source with a documented extract (report name, filters, date), or a Power BI connection if the source supports it. The goal is a repeatable pull, not a fast one.
- **Lay out the page** Status summary at the top, the areas in a fixed order, one visual per metric (a number with trend and threshold color), issues and decisions at the bottom. Same layout every time.
- **Set the refresh and the owner** Who pulls each source, by what day, and who reviews the page before it goes out. Put the as-of date on the page.

REAL TALK

The dashboard is trusted exactly until the first time a number on it is wrong in a meeting. After that, every number is checked against the source and the page becomes something leadership tolerates. The discipline that prevents this is boring: a documented pull, a reviewer who checks the page against the sources before release, and an as-of date. It takes an hour a week and it is the whole difference between a dashboard and a slide.

Spreadsheet or Power BI?

A spreadsheet is enough for most programs: one tab per source, a summary tab with formulas, conditional formatting for thresholds, and sparklines for trend. It is understood by everyone, requires no licensing or IT involvement, and can be built in a day. Power BI or Tableau earns its place when the sources can be connected directly (so the refresh is automatic), when several programs share the same design, or when readers need to drill from the summary into detail. The failure mode with BI tools is building an elaborate model before the metrics are agreed; the failure mode with spreadsheets is a file that only one person understands. Either way, the metrics and the sources are the work; the tool is the presentation.

One sentence to teach your team: **a dashboard is one page of metrics that change decisions, each from a named source, with a threshold set in advance, refreshed on a schedule by a named owner, with the as-of date on it.**

INTERMEDIATE

Designing the Visuals

Each metric gets the smallest visual that carries the information: the current number, a sparkline or arrow for trend over the last six periods, and a color against the threshold. Charts belong only where the shape matters, the burn curve against plan, the S-curve, the milestone timeline. Tables belong for lists: risks, issues, deliverables. Colors are three (on plan, watch, action) with definitions on the page, and they are applied by rule, not judgment. Fonts are one, sizes are two. The page reads top to bottom in the order a reader asks questions: is it healthy, where not, what is being done, what do you need from me. Anything that does not answer one of those is removed.

Leading Indicators: The Part Most Dashboards Miss

Cost variance, milestones met, and defects found are lagging: they report what already happened. A dashboard that shows only lagging indicators tells leadership about problems after they are expensive. Leading indicators predict: open positions against the staffing plan (next quarter's burn shortfall), the trend in risk ratings (next quarter's EAC change), deliverables in government review (next month's schedule), the queue of unpriced changes (next quarter's unbilled). Every area on the dashboard should have at least one leading indicator, and the review should give those the most attention, because they are the ones the program can still act on.

Lagging and Leading Pairs

AREA	LAGGING (WHAT HAPPENED)	LEADING (WHAT IS COMING)
Cost	Cost variance to date	Open positions × planned hours; unpriced change backlog
Schedule	Milestones slipped this period	Critical path float trend; deliverables awaiting government approval
Performance	Availability last month	Open incident trend; component failure rate
Risk	Issues realized this period	Risk rating trend; risks without mitigation owners
Staffing	Positions filled	Clearances in process; time-to-fill trend; attrition notices
Cash	DSO last month	Vouchers rejected this month; unbilled balance trend

Data Lineage: Making Every Number Defensible

For every metric on the page there should be a one-line lineage note in a companion tab: source system, report or query, filters, transformation, and pull date. "Spent to date: accounting system, contract cost report CLIN 0001-0004, through 31 August, sum of direct and indirect." When a reader challenges a number, the lineage is the answer, and when the number is wrong, the lineage is where the error is found. Programs that skip this discover during a dispute or an audit that nobody can explain how a number was produced, which makes every number on the page suspect.

ADVANCED

The Dashboard as a Management Rhythm

A dashboard that is built and emailed is a report. One that is reviewed in a standing meeting, where each red or yellow item has an owner who speaks to it and a decision or action is recorded, is a management system. Senior PMs use the dashboard as the agenda: the review walks the page top to bottom, spends time only on exceptions, records actions in the issues section, and ends with the decisions leadership must make. The page then carries those actions forward to the next review, where they are checked. The dashboard becomes the program's memory of what it said it would do, which is more valuable than any single metric on it.

Dashboards Across Programs and Up the Chain

When several programs use the same layout and definitions, a portfolio view is a matter of stacking the summary rows. That is why standardizing the metric definitions and thresholds across a business unit or a PEO is worth the argument: it makes the portfolio comparable and lets leadership see which programs need attention without reading ten different formats. The costs are that individual programs lose some tailoring, and that a standard threshold may not fit every program's risk profile. The practice is to standardize the core (cost, schedule, staffing, risk) and allow one tailored performance section per program.

Dashboard Failure Modes

Recognizable from across the room.

Everything is green THRESHOLDS

Thresholds set loose, or adjusted to fit, or judgment applied instead of rules.

Leadership stops reading. Reset thresholds from the plan and apply them by rule.

Forty metrics SCOPE

Nothing was cut; the page is a report.

Apply the decision test to each metric. Move the rest to an appendix.

The numbers disagree with the MSR SOURCES

Two products from two pulls of different data.

One source of record per number, used by both. The Monthly Status Report lesson covers the same consistency rule.

No as-of date CURRENCY

Readers do not know how stale the numbers are.

Print it at the top. Every time.

Only one person can refresh it FRAGILITY

A spreadsheet with undocumented pulls and formulas.

Write the pull procedure; have a second person run it once.

WHAT SENIOR PMS DO DIFFERENTLY

They pick metrics that change decisions and cut the rest. They name a source of record for every number and keep the lineage. They set thresholds in advance and apply them by rule. They put leading indicators on the page and spend the review on them. And they run the review from the page, record the actions on it, and check them next time, so the dashboard becomes the program's management system rather than its wallpaper.

BUILD ON THIS

- **Program Metrics & KPIs** Choosing what to measure
- **Data-Driven Decisions & Visualization** Chart selection and presenting to executives
- **Monthly Status Reports** Keeping the dashboard and the MSR consistent

Key Terms

Dashboard

A one-page, regularly refreshed view of a program's status across cost, schedule, performance, risk, and staffing, designed for a reader with five minutes.

Source of Record

The system each number officially comes from: the accounting system for cost, the IMS for schedule, the ticketing system for service metrics, the risk register for risk. Dashboards report from sources of record; they do not become one.

Refresh Cadence

How often each number is updated and by whom. A dashboard is only as current as its slowest input, and readers must know the as-of date.

Threshold

The value at which a metric changes color or triggers action. Set in advance, from the plan or the contract, not adjusted to make the page look better.

Leading vs. Lagging Indicator

Leading indicators predict future performance (open positions, risk trend, backlog of deliverables); lagging indicators report what already happened (cost variance, milestones met). A good dashboard has both.

Data Lineage

The traceable path from a dashboard number back to its source: which report, which field, which date, which transformation. What lets a reader trust the number.

Power BI / Tableau

Business intelligence tools that connect to data sources, transform data, and render visuals with automatic refresh. Widely available in government and contractor environments; not required for a good dashboard.

Sparkline

A small trend line embedded next to a metric, showing direction over recent periods without a full chart. The most information per square inch on a dashboard.

Variance

The difference between plan and actual for any metric. The dashboard shows the variance and, for anything material, the reason.

Single Source of Truth

The principle that one dashboard, from agreed sources, replaces the competing slide decks and spreadsheets that give leadership different numbers for the same program.

Knowledge Check

What is the test for whether a metric belongs on a program dashboard?

- A. Whether it is easy to collect
- ✓ B. Whether a change in it would change a decision leadership would make
- C. Whether the contractor reports it
- D. Whether it is always green

WHY

Metrics that do not inform decisions belong in the appendix or the MSR, not on the page.

What is a "source of record"?

- A. The dashboard itself
- ✓ B. The one system each number officially comes from
- C. The PM's notes
- D. The contractor's slide

WHY

Dashboards report from sources of record; they do not become one. One system per number.

When should thresholds be set?

- A. After seeing the data
- ✓ B. In advance, from the plan and the contract, and applied by rule
- C. At each review by judgment
- D. Never; colors are subjective

WHY

Thresholds adjusted to fit the data produce an all-green page that nobody reads.

Which of these is a leading indicator?

- A. Cost variance to date
- B. Milestones slipped last period

✓ C. Open positions against the staffing plan

- D. Defects found last month

WHY

Open positions predict next quarter's burn shortfall. The others report what already happened.

What is data lineage?

- A. The dashboard's version history

✓ B. The traceable path from a number back to its source, report, filters, transformation, and pull date

- C. The chart type
- D. The refresh schedule

WHY

Lineage is what makes a number defensible when challenged and findable when wrong.

When does Power BI or Tableau earn its place over a spreadsheet?

- A. Always

✓ B. When sources can be connected for automatic refresh, several programs share a design, or readers need drill-down

- C. Never
- D. When the program is small

WHY

The metrics and sources are the work. The tool is the presentation; a spreadsheet is enough for most programs.

What is the most common way a dashboard loses leadership's trust?

A. Too many colors

✓ **B. A number on it is wrong in a meeting, after which every number is checked against the source**

C. It is too short

D. It uses sparklines

WHY

A documented pull, a pre-release reviewer, and an as-of date are the defenses.

What turns a dashboard from a report into a management system?

A. Emailing it weekly

✓ **B. Reviewing it in a standing meeting where exceptions have owners and actions are recorded on the page and checked next time**

C. Adding more charts

D. Printing it in color

WHY

The dashboard becomes the program's memory of what it said it would do.

Why standardize metric definitions across programs in a business unit or PEO?

A. To reduce tailoring

✓ **B. So a portfolio view is a matter of stacking summary rows and leadership can compare programs**

C. Because the FAR requires it

D. To eliminate performance sections

WHY

Standardize the core; allow one tailored performance section per program.

Match the area to its source of record:**CORRECT MATCHES**

Cost	Accounting system (and EVM system if applicable)
Schedule	The integrated master schedule
Risk	The risk register
Deliverables	The deliverable tracker

WHY

Cost from accounting, schedule from the IMS, risk from the register, deliverables from the tracker.

Order the steps to build a dashboard:**CORRECT ORDER**

1. Decide metrics and thresholds with the readers
2. Identify the source of record for each metric
3. Build a documented, repeatable data pull
4. Lay out the page in a fixed order
5. Set the refresh cadence, owner, and reviewer

WHY

Metrics and thresholds with readers, sources of record, repeatable pulls, layout, then refresh and ownership.

A dashboard shows forty metrics, all green, with no as-of date. What is it?

A. A good dashboard

✓ B. A report that nobody reads, with loose thresholds and unknown currency

C. A portfolio view

D. An EVM report

WHY

Three failure modes at once: scope, thresholds, and currency.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 6

Reading a Contractor Cost Report by CLIN

25 min

The monthly cost report is the most detailed financial document a government PM receives and the one most often filed unread. How it is structured, what each column means, how to reconcile it to funding and the invoices, the questions that find problems early, and how a contractor PM should build it so the government trusts it.

The Document Nobody Reads Until It Is Too Late

On a cost-type or T&M contract the contractor sends the government a cost report every month. It shows, by CLIN and often by cost element, what has been spent, what is forecast, and how that compares to what is funded. It is the earliest warning the government gets of an overrun, a funding shortfall, or a scope problem, and it is routinely filed without being read because it looks like an accounting document. This lesson teaches a PM, on either side, to read it in twenty minutes: the structure, the columns, the reconciliations, and the six questions that find problems while they are still cheap.

Anatomy of a Cost Report (CFSR-Style)

COLUMN	WHAT IT MEANS	WHAT TO COMPARE IT TO
CLIN / SLIN	The line item, often with appropriation and fiscal year	The contract's CLIN structure; the funding document
Contract value (ceiling)	The negotiated or ceiling amount for the CLIN	Not the funded amount; do not confuse them
Funds authorized / obligated	What the government has actually put on the CLIN	The government's own obligation records
Incurred to date (actuals)	Cost recorded through the reporting period, usually including accruals	Vouchers submitted; last month's report plus this month's burn
Open commitments	Purchase orders and subcontracts placed but not yet incurred	Subcontract list; material orders
Forecast (ETC) by period	Expected spending in coming months	The staffing plan; the IMS
Estimate at completion	Incurred plus ETC for the CLIN	The CLIN's funded value and ceiling
Funds required / date funds exhausted	When the CLIN runs out at the forecast burn	The funding plan; the Limitation of Funds notice trigger
Variance and explanation	Difference from plan, with the reason	Last month's explanation; the MSR narrative

The Three Numbers That Must Agree

For each CLIN, three numbers come from three places and must reconcile: the cost report's incurred-to-date, the sum of vouchers or invoices submitted through the same date, and the government's record of expenditures against the obligation. They will not match exactly, and the reasons are usually legitimate: accruals in the report that have not been invoiced yet, a voucher in WAWF not yet paid, a rate difference between provisional billing rates and the rates in the report. The discipline is to know the differences and explain them. A gap that grows month over month without explanation means either the report is wrong, the billing is behind, or costs are being incurred that will not be billable. Each of those is a problem the PM wants to find in month three, not month twelve.

The Basic Reconciliation

Incurred to date (cost report)

– Billed to date (vouchers submitted)

= Unbilled (should be explained by accruals, billing lag, and rate differences)

Funded value – Incurred to date – Open commitments = Uncommitted funds remaining

Uncommitted funds remaining ÷ Forecast monthly burn = Months until funds exhausted

The second calculation is the one that triggers Limitation of Funds notices and funding mod requests. If months until exhaustion is less than the funding lead time (often three to four months), the request is already late.

Six Questions That Find Problems Early

- **Is any CLIN forecast to exceed its funded value before the next funding action?** The funds-exhausted date against the funding calendar. This is the Limitation of Funds question and the first thing to check.
- **Is the EAC on any CLIN above its ceiling?** An over-ceiling EAC on a cost-type CLIN is unbillable cost the contractor will eat, which means a scope fight is coming.
- **Did the burn rate change from last month, and does the explanation match the schedule?** A burn that dropped with no schedule change means people left or were moved. A burn that rose with no scope change means something is costing more than planned.
- **Are open commitments consistent with the subcontract and material plan?** Large commitments that appear suddenly mean a subcontract was placed the government did not know about; missing ones mean planned work is not started.
- **Do the cost elements look right?** A labor-heavy CLIN suddenly showing large ODCs, or subcontract cost on a CLIN with no subcontracted scope, is a charging question.
- **Is unbilled growing?** Incurred minus billed rising month over month means work is being done that cannot be invoiced: unpriced changes, funding shortfalls, or rejected vouchers.

REAL TALK

A cost report is the contractor telling the government, in numbers, what is happening on the contract. The government PM who reads it every month and asks one question about it in the next review changes the contractor's behavior: the report gets more careful, the forecasts get more honest, and the surprises get smaller. The PM who never mentions it gets a report that is produced to satisfy a CDRL and nothing more.

Cost Element Detail: What Each Line Tells You

Below the CLIN total, the report usually breaks cost into elements. Direct labor tells you hours and rates by category, which tests the staffing plan. Fringe, overhead, and G&A applied at provisional rates tell you what the government is being billed for indirect cost, and a change in those rates between months means the contractor revised its provisional rates. Subcontracts and materials tell you what was bought and from whom. ODCs (travel, equipment) tell you about activity the SOW should have anticipated. Fee tells you what was earned or billed. A PM who understands the Wrap Rates lesson can read the indirect lines and see, for example, that the overhead rate rose two points, which means the contractor's base shrank and every hour on the contract now costs more. That is worth a question.

One sentence to teach your team: **the cost report tells you by CLIN what was spent, what is committed, what is forecast, and when the money runs out, and the government PM who reconciles it to the vouchers and the funding every month finds overruns and shortfalls while they can still be fixed.**

INTERMEDIATE

Building the Report: The Contractor PM's Side

On the contractor side, the cost report is produced by finance from the accounting system, but the PM owns its accuracy and its forecast. The practice: pull actuals after the monthly close, add accruals for work performed but not yet recorded (subcontractor work in particular), update the ETC by CLIN from the current staffing and subcontract plan (not by copying last month), compare to the funded value and compute the funds-exhausted date, write the variance explanation in the same words the MSR will use, and review the whole thing against the vouchers before it goes out. A report the PM has not reviewed will contradict the MSR, and the government will notice.

Reading Patterns in the Numbers

PATTERN	LIKELY MEANING	QUESTION TO ASK
Burn flat, schedule slipping	Work is taking longer at the same staffing; EAC will rise	What is the revised ETC and why is it not in the report?
Burn rising, scope unchanged	Rework, inefficiency, or unreported scope growth	What changed? Is there an undocumented change?
Burn falling, schedule on track	Efficiency, or work deferred to later periods	Is the ETC still realistic, or is cost being pushed right?
Large accrual reversals	Prior accruals overstated; actuals came in lower	Are accrual estimates reliable? What is the method?
Indirect rates changed between months	Contractor revised provisional billing rates	Was the revision approved by the ACO? What is the impact on EAC?
ETC identical for three months	Forecast is not being updated	When was the ETC last rebuilt from the staffing plan?

Funding by Appropriation and Fiscal Year

On DoD contracts, SLINs often separate funding by appropriation (O&M, RDT&E, procurement) and fiscal year, because each has different rules about what it can pay for and when it expires. The cost report shows incurred and forecast against each. Two problems appear here. Costs charged against a SLIN whose appropriation cannot legally pay for that work (development charged to O&M, for example) is a color-of-money problem for the government. And an expiring appropriation with unspent funds on it is money the government loses if the contractor does not incur against it in time. The government PM reads the report by SLIN with the How DoD Money Works lesson in mind; the contractor PM charges work to the SLIN the government designated for it, not to whichever one has room.

The Report and the Limitation of Funds Notice

The cost report is where the Limitation of Funds (or Limitation of Cost) obligation becomes visible. The clause requires the contractor to notify the CO in writing when it expects costs within a set period (often 60 days) to exceed a set percentage (often 75%) of the funded amount. The report's funds-exhausted date is the trigger. A contractor PM who sees the date inside the notice window and does not send the notice has forfeited the government's obligation to fund the overrun. A government PM who sees the same date and has not started the funding action has a stop-work coming. Both should be reading the same line.

ADVANCED

Cost Reports and EVM: Two Views of the Same Money

On programs with earned value, the cost report and the IPMR draw from the same accounting data and must agree on actuals. The cost report adds the funding view (obligated, funds required, exhaustion date) that the IPMR does not have; the IPMR adds the performance view (earned value, variances, indices) that the cost report does not have. Senior PMs read them together: actual cost in both should match; the IPMR's EAC and the cost report's EAC should match or the difference should be explained (management EAC versus the contractual estimate is a common, legitimate reason); and the cost report's funding picture explains whether the IPMR's forecast can be paid for. The IPMR lesson in this module covers the formats.

Using Cost Reports in Negotiation and Dispute

A year of cost reports is the record of what the contractor knew and said, month by month. In a negotiation over an REA, the reports show when cost growth began and what the contractor attributed it to at the time. In an award fee evaluation, they show whether forecasts were honest. In a dispute over an overrun, they show whether the contractor gave the Limitation of Funds notice when the numbers required it. Senior PMs on both sides keep the reports, read them in sequence, and know that the variance explanation written in month four will be quoted in month sixteen.

A Twenty-Minute Monthly Review

The routine that catches most problems.

Minute 0 to 5: Funding **FUNDS**

For each CLIN: funds-exhausted date versus the next funding action. Anything inside the notice window is the first agenda item.

This is the question with a legal clock on it.

Minute 5 to 10: Totals **EAC**

EAC by CLIN against ceiling and funded value; change from last month; explanation.

An EAC that has not moved in three months is a question, not a comfort.

Minute 10 to 15: Reconciliation **MATCH**

Incurred versus billed; the unbilled trend; the government's expenditure record.

Growing unbilled is the early sign of unpriced work or funding trouble.

Minute 15 to 18: Elements **DETAIL**

Labor hours and mix; indirect rates; subcontracts and ODCs against the plan.

Anything that does not match the SOW's shape gets a question.

Minute 18 to 20: One question ASK

Pick the one item that matters most and put it on the program review agenda.

Asking one good question a month changes the report's quality more than any CDRL language.

WHAT SENIOR PMS DO DIFFERENTLY

They read the cost report every month, funding first. They reconcile it to the vouchers and the obligation records and know the reasons for the differences. They read the cost elements for shape, not just totals. They read the report alongside the IPMR where one exists. And they ask one question about it in every review, because the report the contractor knows will be read is a better report.

BUILD ON THIS

- **Estimate at Completion (EAC)** Building the ETC and the Limitation of Funds clause
- **How DoD Money Works** Appropriations and color of money by SLIN
- **Reading IPMR Reports** The performance view of the same money

Key Terms

Contract Line Item Number (CLIN)

A separately identified and priced item in the contract. The unit of funding, billing, and cost reporting. Sub-CLINs (SLINs) break CLINs down further, often by funding source or fiscal year.

Contract Funds Status Report (CFSR)

The standard DoD cost report (DID DI-MGMT-81468) showing, by CLIN and appropriation, funds authorized, obligated, expended, forecast, and the contractor's estimate of funds needed.

Incurred Cost

Cost the contractor has recorded for work performed, whether or not it has been billed. On a cost-type contract, the basis for the voucher.

Accrued Cost

Cost for work performed that has not yet been recorded in the accounting system (a subcontractor's work not yet invoiced, for example). Accruals make the report reflect reality before the paperwork catches up.

Commitment vs. Obligation vs. Expenditure

On the government side: funds committed (reserved), obligated (on contract), expended (paid). The contractor's report shows expenditures and forecasts against the obligated amount.

Funded Value

The amount actually obligated on the contract or CLIN. Distinct from the contract ceiling or total value. The number the Limitation of Funds clause is measured against.

Estimate to Complete (ETC) and Estimate at Completion (EAC)

ETC is the forecast of remaining cost; EAC is incurred plus ETC. The EAC lesson in the Finance module covers building them; the cost report is where they appear by CLIN.

Cost Element

The category of cost: direct labor, fringe, overhead, subcontracts, materials, ODCs, G&A, fee. Reports break each CLIN down by element.

Burn Rate

Spending per period against the plan. The cost report by month is where burn rate is read directly.

Reconciliation

Confirming that the cost report, the vouchers, and the government's funding records agree, and explaining any differences.

Knowledge Check

What is the difference between a CLIN's contract value and its funded value?

A. None

✓ **B. Contract value is the negotiated or ceiling amount; funded value is what the government has actually obligated**

C. Funded value is always higher

D. Contract value includes fee; funded does not

WHY

The Limitation of Funds clause is measured against the funded amount, not the ceiling.

Which three numbers must reconcile for each CLIN?

A. Ceiling, fee, and profit

✓ **B. The cost report's incurred-to-date, vouchers submitted, and the government's expenditure record**

C. Hours, rates, and headcount

D. EAC, BAC, and PMB

WHY

Differences are usually legitimate (accruals, billing lag, rate differences) but must be known and explained.

Funded value is \$4M, incurred to date \$3.1M, open commitments \$0.3M, forecast burn \$200K per month. When are funds exhausted?

A. In 2 months

✓ **B. In 3 months**

C. In 4.5 months

D. In 6 months

WHY

$(\$4M - \$3.1M - \$0.3M) \div \$0.2M = 3$ months. If funding lead time is 3 to 4 months, the request is already late.

Unbilled (incurred minus billed) is growing every month with no explanation. What does it most likely indicate?

A. Efficient billing

✓ B. Work being performed that cannot be invoiced: unpriced changes, funding shortfalls, or rejected vouchers

C. A lower EAC

D. Fee being earned

WHY

Growing unbilled is the early signal of cash and scope problems.

An EAC on a cost-type CLIN exceeds its ceiling. What does this mean?

A. The government must fund it

✓ B. Cost above the ceiling is unbillable and the contractor will absorb it unless the ceiling is raised; a scope discussion is coming

C. The contract converts to FFP

D. Nothing until closeout

WHY

Over-ceiling cost on cost-type work is a loss unless the government agrees to add ceiling and funds.

The indirect rates applied in the cost report changed between months. What should the PM ask?

A. Nothing; rates always change

✓ B. Whether the contractor revised its provisional billing rates, whether the ACO approved, and what the EAC impact is

C. Whether fee changed

D. Whether the CLIN structure changed

WHY

A provisional rate change raises billed cost on every hour and should be approved and explained.

The ETC by CLIN has been identical for three consecutive months while the schedule slipped. What is happening?

A. Excellent forecasting

✓ B. The forecast is not being rebuilt from the staffing plan; the EAC is stale

C. The contract is under budget

D. Accruals are reversing

WHY

A flat ETC with a slipping schedule is a forecast nobody is updating.

Why are DoD CLINs often split into SLINs by appropriation and fiscal year?

A. For convenience

✓ B. Because each appropriation has different rules about what it can pay for and when it expires

C. To increase fee

D. To satisfy DCAA

WHY

Charging work to the wrong appropriation is a color-of-money violation; unspent expiring funds are lost.

The cost report's funds-exhausted date falls inside the Limitation of Funds notice window. What must the contractor do?

A. Nothing until funds run out

✓ B. Send the written notice to the CO as the clause requires, or forfeit the government's obligation to fund the overrun

C. Stop work immediately

D. Bill the ceiling

WHY

The notice is a contractual obligation with a clock. The report is where the trigger is visible.

Match the pattern to its likely meaning:**CORRECT MATCHES**

Burn flat, schedule slipping	Work taking longer; EAC will rise
Burn rising, scope unchanged	Rework or undocumented scope growth
ETC unchanged for three months	Forecast not being updated
Unbilled growing monthly	Unpriced work or funding trouble

WHY

Flat burn with slipping schedule means EAC will rise; rising burn with unchanged scope means rework or unreported growth; flat ETC means stale forecasting; growing unbilled means unbillable work.

Order the twenty-minute monthly review:**CORRECT ORDER**

1. Funds-exhausted date per CLIN versus the funding calendar
2. EAC by CLIN against ceiling and funded value
3. Reconcile incurred to billed and to the obligation record
4. Read cost elements for shape against the SOW
5. Pick one question for the program review

WHY

Funding first (it has a legal clock), then EAC totals, then reconciliation, then cost elements, then one question for the review.

On a program with EVM, how should the cost report and the IPMR relate?

A. They are unrelated

✓ B. Actuals must match; EACs should match or the difference be explained; the cost report adds the funding view and the IPMR adds the performance view

C. The IPMR replaces the cost report

D. The cost report replaces the IPMR

WHY

Two views of the same money, read together.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 7

Forecasting Burn: Simple Models That Beat the Straight Line

24 min

Most programs forecast spending by drawing a straight line through last month. This lesson replaces that with three methods a PM can run in a spreadsheet: staffing-driven forecasts, trend with seasonality, and S-curve fitting. When each one applies, how to build it, how to check it against actuals, and how to turn the forecast into funding requests and EAC updates that hold up.

Why the Straight Line Is Always Wrong

Ask most programs how they forecast spending and the answer is some version of "last month times the months remaining." It is fast and it is wrong in every direction: it ignores the two people starting next month, the sub whose invoice comes quarterly, the December holidays, the surge planned for the test event, and the fact that the last month was itself abnormal. The result is funding requests that are late, ETCs that surprise finance, and funds-exhausted dates that arrive early. This lesson gives three methods that take an afternoon to set up and a few minutes a month to maintain, and it explains which one fits which kind of work.

Three Methods

METHOD	HOW IT WORKS	BEST FOR	EFFORT
Staffing-driven	Planned headcount by category by month × available hours × burdened rate, plus subcontracts and ODCs by planned timing	Services, level-of-effort, T&M; anything where labor is the driver	An afternoon to build; minutes monthly
Trend with seasonality	Recent-period average adjusted for known monthly patterns (holidays, quarterly sub invoices, fiscal-year surges)	Stable recurring work with a history; a cross-check for the other methods	An hour
S-curve fit	Fit the cumulative actuals to a standard S-curve and project the remainder to the EAC	Development, production, construction; efforts with a defined end	An hour with a template; must be re-fit as actuals accrue

The Staffing-Driven Forecast, Step by Step

Start with the staffing plan: for each labor category, the number of people expected on the contract in each future month, including planned starts, planned departures, and known leave. Multiply by that month's available hours (a calendar of paid hours minus holidays, and an assumed leave rate, so December might be 140 hours and March 170). Multiply by the burdened rate for the category, using provisional rates on cost-

type work or contract rates on T&M. Add subcontracts by their expected invoice timing, not smoothed. Add ODCs by planned events (travel for the test, equipment for the install). Sum by month. The result is a forecast that moves when the drivers move: a delayed hire pushes the burn right, a holiday month drops it, a sub's quarterly invoice spikes it. That is what actual spending does, which is why this method is accurate.

The Staffing-Driven Forecast

Monthly labor forecast = $\sum$ over categories (Headcount $\times$ Available hours that month $\times$ Burdened rate)

Monthly forecast = Labor + Subcontract invoices due + ODCs by event

Funds-exhausted month = First month where cumulative (actuals + forecast) $\geq$ funded value

A 12-person team at an average \$150 burdened rate: $12 \times 168 \text{ hours} \times \$150 = \$302\text{K}$ in a full month, but $12 \times 136 \times \$150 = \245K in December. A straight line at \$302K overstates December by \$57K and understates the funds-exhausted date by weeks.

The Inputs That Move the Forecast Most

- **Start dates** A hire forecast for the 1st who starts on the 20th is two-thirds of a month of burn that did not happen. Forecast from the realistic start, not the requisition date.
- **Available hours by month** Build a calendar. Holiday-heavy months run 10% to 20% lighter.
- **Subcontract invoice timing** Monthly, quarterly, or milestone-based. Smoothing quarterly invoices into monthly burn hides a spike the funding has to cover.
- **Rate changes** Provisional rate revisions, option-year escalation, wage determination updates. Each changes every hour after its effective date.
- **Planned events** Test campaigns, deployments, surges, and their ODCs. Put them in the month they happen.
- **Attrition** A team of 12 with 15% annual attrition loses about one person every seven months; forecasting full staffing all year overstates burn.

REAL TALK

The forecast is only as good as the staffing plan behind it, and the staffing plan is only honest if the PM updates it when reality changes. The person who was going to start in March and now starts in May, the sub who is a month behind, the surge that got cancelled: each one goes into the model the week it is known. A forecast rebuilt monthly from a current staffing plan is right within a few percent. One built once and extended is a straight line with extra steps.

Turning the Forecast Into Actions

The forecast produces three outputs the PM uses every month. The ETC by CLIN, for the cost report and the EAC review, is the sum of the forecast for the remaining periods. The funds-exhausted date, for the CO and the resource manager, is where cumulative actuals plus forecast reach the funded value; subtract the funding lead time and that is the date the funding request must be submitted. And the burn versus plan comparison, for the dashboard and the MSR, is the forecast against the original spend plan by month, with the reasons for the difference. A PM who can produce these three from one model every month is ahead of most.

One sentence to teach your team: **forecast spending from the drivers, headcount by month times real available hours times rate plus subcontracts and events when they actually occur, rebuild it monthly, and read off the ETC, the funds-exhausted date, and the variance from plan.**

INTERMEDIATE

The S-Curve for Efforts With a Shape

Development, production, and construction efforts do not burn evenly. They start slowly (planning, staffing, design), accelerate through the middle (build, integrate, test), and taper at the end (closeout, acceptance). Cumulative spending traces an S. Given actuals to date and an EAC, an S-curve fit projects the remaining months by finding where on the curve the program is and following the shape. A simple version: express cumulative actuals as a percentage of EAC, find the corresponding point on a standard S-curve (a logistic function, or the program's own planned curve from the baseline), and project the remainder along it. The method is strong when the EAC is credible and the effort really has the shape; it is misleading when the effort is level-of-effort or when the EAC is stale. Re-fit monthly, and compare the fitted curve to the staffing-driven forecast; where they disagree, one of the inputs is wrong.

Trend With Seasonality: The Cross-Check

For stable recurring work, a trend forecast is a fair approximation and a good check on the staffing model. Take the last six to twelve months of actuals, compute a monthly average, and adjust each future month by a seasonality factor derived from the program's own history or the available-hours calendar (December at 0.85, March at 1.02, and so on). Add known non-recurring items. If the trend forecast and the staffing forecast differ by more than 10% for a month, find out why: usually the staffing plan has a start or departure the trend does not know about, or the trend includes an abnormal month. The disagreement is information.

Forecast Accuracy: Tracking It

PERIOD	FORECAST (MADE 1 MONTH PRIOR)	ACTUAL	ERROR	CAUSE
March	\$298K	\$305K	+2%	Overtime on test prep
April	\$310K	\$262K	-15%	Two hires slipped to May; sub invoice late
May	\$330K	\$351K	+6%	Late sub invoice landed; hires started
June	\$318K	\$322K	+1%	On plan
Rolling 3-month			-3%	Timing, not total; cumulative within 3%

Reporting a Range, Not Just a Number

A forecast has uncertainty and the reader should see it. Build a low and a high case by flexing the two or three inputs that matter most: hire dates (early and late), a sub's timing, an event that might slip. Report the point forecast with the range: "ETC \$2.4M, range \$2.2M to \$2.7M; the high case assumes the surge proceeds and the March hire starts on time." Finance can reserve against the range; the CO can plan funding against the high case; leadership knows what could change the number. A single point forecast invites the question "how confident are you," and the honest answer should already be on the page.

ADVANCED

Forecasting Across a Portfolio

At the business unit or PEO level, program forecasts roll up into the direct labor base for the rate build (the Indirect Rate Games lesson in the Business module), into the funding plan by appropriation, and into the revenue forecast. Rolled-up straight lines produce a rate build that is wrong, funding requests that bunch in the fourth quarter, and revenue forecasts that miss. Rolled-up staffing-driven forecasts produce a base that reflects real hiring plans and real seasonality. Senior leaders who require each program to forecast from drivers, in a common template, get a portfolio forecast they can use. Those who accept straight lines get a sum of guesses.

When the Forecast Says the Money Runs Out

The forecast's most important output is bad news delivered early: the funds-exhausted date is before the next funding action, or the ETC exceeds the ceiling. The senior PM's response is a sequence, not a scramble. Verify the forecast (a re-run with current inputs). Quantify the gap and the date. Identify options: accelerate funding, slow the burn (defer a hire, delay an event), descope, or request added ceiling. Notify:

the Limitation of Funds notice if the clause requires it, the CO and resource manager regardless, finance for the EAC. And document the options and the decision. The forecast that surfaces the problem four months out gives all of those options time to work. The straight line that surfaces it four weeks out gives one option: stop.

Building the Model: A Spreadsheet Layout That Lasts

One workbook, five tabs, maintained monthly.

Calendar INPUTS

Available hours by month for the period of performance: paid hours minus holidays minus assumed leave.

Built once; the source of the seasonality everything else uses.

Rates INPUTS

Burdened rate by labor category with effective dates for changes.

Provisional rates on cost-type; contract rates on T&M; escalation and WD updates at options.

Staffing plan DRIVERS

Headcount by category by month, with start and end dates per person or position.

The tab the PM edits monthly. Every change here flows through.

Subs and ODCs DRIVERS

Subcontract invoices by expected month; ODCs by event.

Not smoothed. In the month they occur.

Forecast and accuracy OUTPUTS

Monthly forecast, cumulative, funded value line, funds-exhausted date, ETC by CLIN, and the forecast-versus-actual log.

The tab leadership sees. The accuracy log is what makes it credible.

WHAT SENIOR PMS DO DIFFERENTLY

They forecast from drivers, not from last month. They rebuild the staffing plan the week reality changes. They keep a calendar of real available hours. They put subcontract invoices and events in the months they happen. They report a range with the point. They track accuracy and know their error. And when the model says the money runs out, they act on it with months to spare.

BUILD ON THIS

- **CPAF, Burn Rate & Award Fee** Burn rate as the speed limit nobody posts
- **Unit Economics of a Billable Hour** Available hours and utilization
- **Reading a Contractor Cost Report by CLIN** Where the forecast is reported

Key Terms**Burn Forecast**

A projection of spending by period for the remainder of a contract or CLIN. The basis for funding requests, the ETC, and the funds-exhausted date.

Straight-Line Forecast

Projecting future periods at the average of recent periods. Simple, wrong whenever staffing, scope, or the schedule changes, which is most of the time.

Staffing-Driven Forecast

Building the forecast from the planned headcount by labor category, by month, times hours and burdened rates, plus planned subcontracts and ODCs. The most accurate method for services because it models the drivers.

Available Hours

Paid hours in a month minus holidays and expected leave. Varies by month (December and July are short); a forecast that uses a flat monthly figure is wrong by up to 15% in some months.

S-Curve

The cumulative spending shape of most development and production efforts: slow start, steep middle, tapering end. Fitting an S-curve to actuals to date projects the remainder.

Run Rate

Current spending per period, annualized or extended. A useful check, not a forecast.

Seasonality

Recurring patterns by month: fewer hours in holiday months, year-end surges in ODCs, fiscal-year effects on government-directed work.

Forecast Accuracy

The comparison of what was forecast for a period to what actually happened, tracked over time. The only way to know whether the method works.

Confidence Range

A low and high case around the forecast, driven by the main uncertainties (staffing timing, scope, rate changes). Reported alongside the point forecast.

Funds-Exhausted Date

The date the forecast reaches the funded value. The output that matters most to the CO and the resource manager.

Knowledge Check

Why is a straight-line burn forecast usually wrong?

- A. It is too complex
- ✓ B. It ignores planned staffing changes, subcontract invoice timing, holiday months, and planned events
- C. It requires a data team
- D. It is only for T&M

WHY

Actual spending moves with its drivers; a straight line models none of them.

A 12-person team at \$150 burdened: 168 available hours in a normal month, 136 in December. What is the December forecast?

- A. \$302K
- ✓ B. \$245K
- C. \$180K
- D. \$320K

WHY

$12 \times 136 \times \$150 = \$244,800$. A straight line at \$302K overstates December by about \$57K.

Which forecasting method is best for a services contract where labor is the main driver?

- A. S-curve fit
- ✓ B. Staffing-driven: headcount by category by month $\times$ available hours $\times$ burdened rate, plus subs and ODCs by timing
- C. Run rate
- D. Straight line

WHY

Modeling the drivers is what makes the forecast move when reality moves.

When does an S-curve fit apply well?

A. Level-of-effort support

✓ B. Development, production, or construction efforts with a defined end and a credible EAC

C. Any contract

D. Only fixed-price work

WHY

S-curves model efforts with a slow-fast-slow shape. They mislead on level-of-effort work or with a stale EAC.

How should quarterly subcontractor invoices be treated in the forecast?

A. Smoothed evenly across months

✓ B. Placed in the months they are expected, unsmoothed, because funding must cover the spike

C. Excluded

D. Annualized

WHY

Smoothing hides a spike the funded value has to absorb in one month.

The staffing forecast and the trend forecast differ by 15% for a month. What does that mean?

A. Use the average

✓ B. One input is wrong; find out which, usually a start or departure the trend does not know about or an abnormal month in the trend

C. Both are wrong

D. Ignore the trend

WHY

The disagreement is information. Reconcile it.

What is the funds-exhausted date?

A. The contract end date

✓ B. The first month where cumulative actuals plus forecast reach the funded value

C. The date the ceiling is reached

D. The fiscal year end

WHY

Subtract the funding lead time to get the date the funding request must be submitted.

Why report a forecast range rather than a single number?

A. To avoid accountability

✓ B. So finance can reserve, the CO can plan funding against the high case, and leadership sees what could change the number

C. Because the FAR requires it

D. To make the dashboard longer

WHY

Flex the two or three inputs that matter most; report low, point, and high with the assumptions.

What is the purpose of a forecast accuracy log?

A. Compliance

✓ B. To compare each period's forecast to its actual over time, so the PM knows the method's error and where it comes from

C. To replace the cost report

D. To calculate fee

WHY

Tracking accuracy is the only way to know whether the method works and to make the forecast credible.

Match the method to its best use:**CORRECT MATCHES**

Staffing-driven	Services and level-of-effort where labor drives cost
S-curve fit	Development or production with a defined end
Trend with seasonality	Stable recurring work; cross-check

WHY

Staffing-driven for labor-driven services; S-curve for shaped efforts; trend with seasonality as a cross-check on stable recurring work.

Order the response when the forecast shows funds running out before the next funding action:**CORRECT ORDER**

1. Re-run the forecast with current inputs to verify
2. Quantify the gap and the date
3. Identify options: accelerate funding, slow burn, descope, add ceiling
4. Notify the CO and resource manager (and send the LOF notice if required)
5. Document the options and the decision

WHY

Verify, quantify, identify options, notify, document.

At the portfolio level, why do rolled-up straight-line forecasts cause problems?

- A. They do not
- ✓ **B. They produce a wrong direct labor base for the rate build, bunched funding requests, and missed revenue forecasts**
- C. They are too accurate
- D. They require Power BI

WHY

Portfolio forecasts are only as good as the program forecasts beneath them.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 8

Contract Data as Intelligence: FPDS, USASpending, and SAM.gov

24 min

The government publishes every contract award it makes. Almost nobody uses the data. How the three main sources fit together, what each one can and cannot tell you, the questions a capture team, a government PM, or a competitor analyst can answer in an afternoon, and the traps in the data that produce confident wrong answers.

The Government Tells You Everything It Bought

Every federal contract action above the micro-purchase threshold is recorded and published: who bought what from whom, for how much, under what vehicle, competed how, set aside for whom. The record goes back decades. It is free. And in most capture teams and program offices it is used, if at all, for a one-time look at a competitor. This lesson treats it as what it is: a complete history of the market a company sells into or a program office buys from. The DoD Contracting Enterprise lesson introduced SAM.gov and USASpending as intelligence feeds. This lesson goes deeper on the three sources, the questions they answer, and the ways the data misleads people who do not know its structure.

Three Sources, Three Jobs

SOURCE	WHAT IT CONTAINS	BEST FOR	LIMITS
SAM.gov contract opportunities	Sources sought, RFIs, solicitations, amendments, award notices, J&As; entity registrations	What is coming and what was just awarded; who is registered under which NAICS	Only what COs post; many orders under vehicles are not posted; award notices are inconsistent
USASpending.gov	FPDS contract data presented searchably, with downloads; plus grants and assistance	Fast searches and downloads by agency, vendor, NAICS, PSC, place, time	Presents FPDS data with the same lag and errors; some fields simplified
FPDS (via its own reports or the full dataset)	The complete contract action record with every field	Detailed analysis: competition, contract type, vehicle, office, modifications over time	Harder interface; data quality varies by field and agency

What a Contract Action Record Contains

Each FPDS record is one action on one contract. The fields that matter for analysis: the contracting office and the funding office (who bought it and who paid); the vendor and its size and socioeconomic status at

award; the NAICS and PSC (the industry and the product or service); the obligated amount for this action and the base-and-all-options value of the contract; the contract type and pricing arrangement; the extent competed and, if not fully competed, the reason; the set-aside type; the referenced vehicle if it is an order; the period of performance; the place of performance; and a short description. A contract's history is the set of all its actions: the award, then each modification and funding action over its life. Reading the history tells you how the contract grew, when options were exercised, and when it ended.

Questions an Afternoon Can Answer

- **Who buys what I sell, and where?** Search by PSC and NAICS, group by contracting office. The offices with recurring awards in your category are your customers, whether or not they know you.
- **Who wins there?** For those offices, group by vendor. The share held by each, the small business share, the incumbents on the largest contracts.
- **How is it bought?** Extent competed, set-aside type, contract type, and referenced vehicle. If most awards are orders under one vehicle, you need to be on that vehicle. If most are set aside for a category you hold, that is your door.
- **When do the big ones end?** Period of performance end dates on the largest contracts in your category. Those are the recompetes, and the timeline lesson says the government starts them two years out.
- **What does it cost?** Obligated dollars over the life of comparable contracts, divided by scope from the description or the solicitation. The basis for an IGCE or a price-to-win estimate.
- **What did the competitor just win?** Vendor search, sorted by date. Their new awards, their growth, their customer concentration, their vehicle positions.

REAL TALK

The most valuable single search a small contractor can run is: my PSC codes, my target agency, last three years, grouped by contracting office and vendor. It shows which offices buy the work, who they buy it from, whether they set it aside, and which contracts are ending. Most companies that run it for the first time discover an office they have never called that has awarded their kind of work a dozen times.

The Traps

The data is comprehensive and it is also full of ways to be wrong. Obligated dollars on one action are not the contract value; sum the actions or read the base-and-all-options field. A modification with a large obligation may be a funding action on an existing contract, not a new award. Contract type and competition fields are sometimes miscoded. NAICS codes are chosen by the CO and vary for the same work; search by PSC as well. Orders under vehicles show the vehicle holder as vendor, which can hide a sub doing the work. Vendor names and identifiers change with mergers. And there is lag: recent months are incomplete. Every conclusion from the data should be checked against a second source (the solicitation on SAM.gov, the vendor's own announcements, a conversation with the customer) before it drives a decision.

One sentence to teach your team: **every award the government makes is public in FPDS and USASpending, so who buys your kind of work, from whom, how, and when it ends is an afternoon of searching away, as long as you sum the actions, search by PSC as well as NAICS, and check the conclusions against a second source.**

INTERMEDIATE

Building a Customer Map From the Data

The DoD Contracting Enterprise lesson described building a customer map. The data makes it systematic. Pull three years of actions in your PSCs and NAICS codes for your target department. Group by contracting office: total dollars, number of awards, small business share, dominant vehicles. For each office with meaningful activity, list the largest contracts with their vendors, values, end dates, and competition type. Add the office's small business specialist and its published forecast. The result is a one-page profile per office: what it buys, how, from whom, and when the next opportunities fall. Refresh it quarterly. This is the capture pipeline's foundation, built from public records instead of guesswork.

Competitor Analysis: Reading a Company Through Its Awards

A vendor's FPDS history is a partial financial statement. Total obligations by year show growth or decline. Customer concentration shows dependence. The set-aside share shows how much of the business depends on small status. The vehicle positions show where it can compete. The end dates of its largest contracts show its recompute exposure. The competition fields show whether it wins full and open or lives on sole source and set-asides. For a public company, the Reading a 10-K lesson in the Business module gives the whole picture; for a private one, FPDS is most of what is available. Capture teams use it for black hat reviews; buyers use it in diligence; a government PM uses it to check a bidder's claimed past performance against what the record shows.

Field Guide to the Fields That Mislead

FIELD	WHAT PEOPLE ASSUME	WHAT IT ACTUALLY IS	FIX
Obligated amount	The contract's value	This action's obligation only	Sum actions on the contract, or use base-and-all-options value
Vendor	The company doing the work	The company holding the contract or vehicle	For orders, check the referenced vehicle; consider subs
NAICS	A consistent category	The CO's choice for that solicitation	Search by PSC as well; use several NAICS codes
Extent competed	Reliable	Often miscoded, especially for orders	Cross-check with the solicitation on SAM.gov
Date signed	Award date	The action's signature date; awards can post weeks later	Allow for lag; do not treat the last two months as complete
Description	What was bought	A short free-text field of variable quality	Read the solicitation for scope

For Government PMs: Market Research From the Record

The Market Research lesson in the Pre-Award module lists database research as the starting technique. This is how to do it. For a new requirement, search the PSC and NAICS government-wide: who has bought this, from which vendors, on which vehicles, at what values, competed how. That shows whether small businesses have won comparable work (evidence for the Rule of Two), what contract types the market has accepted, what comparable contracts cost (evidence for the IGCE), and which vehicles already hold capable firms. Document the search in the market research report with dates and filters. A market research report that cites specific FPDS findings is far more defensible than one that says "a search was conducted."

ADVANCED

Working With the Full Dataset

For recurring analysis, downloading the data beats searching it. USASpending offers bulk downloads by agency and period, and FPDS offers reports and an API. A capture analyst or a program office analyst pulls the relevant slice (a department, three years, the PSCs of interest) into a spreadsheet or a database, cleans vendor names, sums actions by contract, tags contracts by end date, and builds the office and vendor profiles from there. Refreshed quarterly, this becomes a living market model: pipeline candidates appear as contracts approaching end dates, competitor moves appear as new awards, and set-aside patterns appear as trends. The Dashboard lesson's discipline applies: documented pulls, named sources, an as-of date.

Combining Sources: The Full Picture of One Opportunity

For a specific recompetete, the three sources together tell the story. FPDS gives the incumbent contract: award date, value, modifications, options exercised, contract type, competition, vehicle, and the end date. SAM.gov gives the original solicitation (scope, Section L and M, the NAICS, the set-aside), the award notice, any J&As on bridges, and, when the recompetete starts, the sources sought. USASpending gives the incumbent's wider profile and the office's buying pattern. Add the office's published forecast and a conversation with the small business specialist, and the capture team knows the requirement, the incumbent, the customer's habits, the likely acquisition strategy, and the timing, before the first sources sought is posted. That is what shaping-window positioning is built on.

A Quarterly Market Intelligence Routine

For a small capture team, a few hours a quarter.

Refresh the pull DATA

Download the last quarter's actions in your PSCs and NAICS codes for your target departments; append to the model.

Same filters every time, documented.

Update office profiles CUSTOMERS

New awards by office; changes in set-aside share, vehicles, and contract types.

An office that started setting work aside this year is a new door.

Flag ending contracts PIPELINE

Contracts in your categories ending in 18 to 30 months; add to the pipeline as recompetete candidates.

Then watch SAM.gov for the sources sought.

Track competitors COMPETITION

New awards, lost recompetetes, vehicle wins, growth trend for the five firms you compete with most.

Feeds the next black hat review.

Cross-check anything actionable VERIFY

Before a bid decision or a customer call, confirm the data against SAM.gov and a human source.

The data finds the question. It does not answer it alone.

WHAT SENIOR PMS AND CAPTURE LEADS DO DIFFERENTLY

They build the customer map from the record, not from memory. They sum actions and search by PSC. They know the fields that mislead and cross-check them. They refresh the model quarterly and let ending contracts populate the pipeline. And they treat FPDS as the first step of market research on the government side and the first step of capture on the contractor side, because it is the same data and the same questions.

BUILD ON THIS

- **The DoD Contracting Enterprise** SAM.gov and USASpending introduced; building a customer map
- **Market Research** The government's use of the same data
- **The BD-to-Capture Lifecycle** Where the intelligence feeds the pipeline

Key Terms

Federal Procurement Data System (FPDS)

The government's system of record for contract actions above the micro-purchase threshold: every award, modification, and order, with agency, vendor, NAICS, PSC, dollars, contract type, competition, set-aside, and more. Reported by contracting officers.

USASpending.gov

The public portal that presents federal spending, including FPDS contract data, with a friendlier interface, downloadable datasets, and links to grants and other assistance. The easiest way in.

SAM.gov

Where solicitations, sources sought, award notices, and J&As are posted (contract opportunities), and where entities register. The forward-looking source.

Product Service Code (PSC)

A four-character code describing what was bought (a service category or a product class). Complements the NAICS, which describes the industry. Searching by PSC finds the work regardless of which NAICS the CO chose.

Contract Action

A single FPDS record: an award, a modification, or an order. A contract's total value is the sum of its actions; a single action is not the contract.

Obligated Amount

The dollars put on the contract by that action. Distinct from the contract's ceiling or total potential value, which FPDS reports separately (base and all options value).

Extent Competed

The FPDS field recording whether the award was full and open, set aside, or not competed, with the exception cited. The field that reveals sole-source patterns.

Referenced IDV

For an order, the indefinite delivery vehicle it was placed under. Lets you trace orders back to their vehicle and see which vehicles a customer actually uses.

Contracting Office

The specific office that made the award, identified by code. The unit of analysis for "who buys this and where."

Data Lag and Quality

FPDS records are entered by COs after award, with delays and errors. The data is comprehensive but not clean; conclusions need cross-checks.

Knowledge Check

What is FPDS?

A. A forecasting tool

✓ **B. The government's system of record for contract actions above the micro-purchase threshold, reported by contracting officers**

C. The SBA's size database

D. A pricing schedule

WHY

Every award, modification, and order is a record with agency, vendor, NAICS, PSC, dollars, type, competition, and more.

How does USASpending.gov relate to FPDS?

A. They are unrelated

✓ **B. USASpending presents FPDS contract data through a searchable, downloadable interface, along with grants and other assistance**

C. USASpending replaced FPDS

D. FPDS is for grants only

WHY

Same underlying data, friendlier access, same lag and errors.

Why search by Product Service Code as well as NAICS?

A. PSC is more accurate

✓ **B. The NAICS is the CO's choice and varies for the same work; PSC describes what was bought and finds the work regardless**

C. NAICS is not in FPDS

D. PSC includes price

WHY

NAICS describes the industry; PSC describes the product or service. Use both.

A single FPDS record shows a \$12M obligation. What is it?

A. The contract's total value

✓ **B. One action's obligation; it may be an award or a funding modification, and the contract's value is the sum of its actions**

C. The annual value

D. The ceiling

WHY

Sum the actions or read the base-and-all-options field. A large modification is often a funding action, not a new award.

Which field reveals sole-source patterns at a contracting office?

A. Place of performance

✓ **B. Extent competed, with the exception cited when not fully competed**

C. PSC

D. Vendor name

WHY

Extent competed records full and open, set-aside, or not competed with the reason, though it is sometimes miscoded.

For an order under a vehicle, who appears as the vendor in FPDS?

A. The subcontractor doing the work

✓ **B. The vehicle holder that received the order, which can hide a sub doing the work**

C. The contracting officer

D. The vehicle sponsor

WHY

Check the referenced IDV and consider that subs are invisible in the record.

What is the most valuable single search for a small contractor?

A. All federal spending

✓ B. Its PSC codes, its target agency, last three years, grouped by contracting office and vendor

C. Its own awards

D. Grants

WHY

It shows which offices buy the work, from whom, whether set aside, and which contracts are ending.

How does a government PM use FPDS in market research?

A. To award contracts

✓ B. To find who has bought comparable work, from which vendors, on which vehicles, at what cost, competed how, as evidence for the Rule of Two, contract type, and the IGCE

C. To post solicitations

D. To register vendors

WHY

A market research report citing specific FPDS findings is far more defensible than one that says a search was conducted.

Why should conclusions from FPDS be cross-checked?

A. They should not

✓ B. The data has lag, miscoded fields, inconsistent NAICS, and vendor name changes; a second source confirms before a decision

C. FPDS is classified

D. Only USASpending is reliable

WHY

The data finds the question. SAM.gov, vendor announcements, and human sources answer it.

Match the source to its best use:**CORRECT MATCHES**

SAM.gov contract opportunities	Sources sought, solicitations, award notices, J&As
USASpending.gov	Fast searches and bulk downloads of award data
FPDS full record	Detailed analysis of competition, type, vehicle, and modifications

WHY

SAM.gov for what is coming and just awarded; USASpending for fast searches and downloads; FPDS for detailed field-level analysis.

Order the quarterly market intelligence routine:**CORRECT ORDER**

1. Refresh the data pull with the same documented filters
2. Update contracting office profiles
3. Flag contracts ending in 18 to 30 months as pipeline candidates
4. Track the five closest competitors' awards and losses
5. Cross-check anything actionable against SAM.gov and a human source

WHY

Refresh the pull, update office profiles, flag ending contracts, track competitors, cross-check before acting.

For a specific recompet, what does combining FPDS, SAM.gov, and USASpending give a capture team?

A. The winning price

✓ B. The incumbent contract's history, the original solicitation and any bridges, the incumbent's profile, and the office's buying pattern, before the sources sought is posted

C. The evaluation scores

D. The IGCE

WHY

That is what shaping-window positioning is built on.

DATA ANALYTICS FOR PROGRAM MANAGERS · MODULE ASSESSMENT

Module Assessment

Pass this to unlock the next skill level in the app.

Knowledge Check

In DoD program reporting, what does the acronym "EVM" stand for and what does it measure?

- A. Extended Value Metrics: contractor profitability
- ✓ **B. Earned Value Management: integrates cost, schedule, and technical performance into a single management framework**
- C. Estimated Variance from Milestone: deviation from program schedule
- D. Enterprise Visibility Mechanism: contractor data reporting system

WHY

Earned Value Management (EVM) is a project management technique that integrates scope, schedule, and cost. It compares planned work (PV), work accomplished (EV), and actual cost (AC) to provide objective performance indicators like CPI and SPI. Required on contracts over \$50M (raised from \$20M in Feb 2026 under DFARS Class Deviation 2026-O0011, part of the DFARS overhaul).

A Selected Acquisition Report (SAR) is submitted to Congress when a program exceeds cost growth thresholds. What is this threshold called?

- ✓ **A. Nunn-McCurdy Breach**
- B. Anti-Deficiency Violation
- C. Should-Cost Exceedance
- D. Independent Cost Estimate Variance

WHY

The Nunn-McCurdy Act requires DoD to notify Congress when a program's unit cost growth exceeds 15% (significant) or 25% (critical) above the current baseline, or 30%/50% above the original baseline. A critical breach requires the SecDef to certify the program is essential to national security or terminate it.

In data analytics for DoD programs, what does "CDRL" stand for and why does it matter?

✓ **A. Contract Data Requirements List: defines the data deliverables the contractor must provide the government under the contract**

B. Contractor Deficiency Resolution Log: tracks contractor performance issues

C. Cost Data Reporting Level: defines EVM reporting thresholds

D. Consolidated Defense Reporting List: OSD-level program status reports

WHY

A CDRL (Contract Data Requirements List, DD Form 1423) specifies every data item the government requires the contractor to deliver. Technical reports, schedules, cost reports, test results. CDRLs are legally binding contract deliverables. Missing or late CDRLs are a contract performance issue the COR must document.

What is the primary purpose of a Defense Acquisition Executive Summary (DAES)?

A. To certify program funding at milestone reviews

✓ **B. To provide OSD leadership with a monthly assessment of program health: cost, schedule, and performance: using color-coded ratings**

C. To document contractor past performance for future source selections

D. To report financial audit results to the Inspector General

WHY

The DAES is a monthly program health report reviewed at the OSD level. It uses a color-coded assessment (Green/Yellow/Red) across cost, schedule, and performance dimensions. Programs in amber or red status receive additional OSD scrutiny and may be required to present recovery plans.

Which metric measures the cost efficiency of work completed to date in EVM?

A. SPI (Schedule Performance Index)

B. TCPI (To-Complete Performance Index)

✓ **C. CPI (Cost Performance Index)**

D. VAC (Variance at Completion)

WHY

$CPI = EV / AC$. It measures how much earned value you are getting per dollar spent. A CPI of 1.0 means on budget; below 1.0 means cost overrun; above 1.0 means under budget. CPI tends to stabilize early in a program and is a reliable predictor of final cost performance.

A program's Integrated Master Schedule (IMS) shows the critical path. What does "critical path" mean?

A. The tasks with the highest dollar value in the contract

✓ **B. The longest sequence of dependent tasks that determines the minimum program completion date: any slip on the critical path slips the whole program**

C. The tasks most likely to experience technical failure

D. The top 10 highest-risk tasks identified in the program risk register

WHY

The critical path is the longest chain of logically dependent activities in the schedule. Tasks on the critical path have zero float. Any delay directly extends the program end date. PMs must track critical path tasks with the most scrutiny. A schedule with no margin on the critical path is a high-risk program.

What does "FPRR" stand for and who conducts it?

A. Fixed Price Rate Review: conducted by DCMA to review labor rates at contract award

✓ **B. Forward Pricing Rate Recommendation: issued by DCAA based on audit of contractor indirect cost rates used in pricing future proposals**

C. Final Program Risk Review: conducted by the PM at Milestone C

D. Federal Procurement Reporting Requirement: annual OSD data call

WHY

DCAA issues Forward Pricing Rate Recommendations (FPRRs). Audit-based assessments of a contractor's proposed indirect cost rates (overhead, G&A, fringe). These rates are used in pricing cost-type contracts and T&M proposals. A CO may use the FPRR or negotiate Forward Pricing Rate Agreements (FPRAs) directly with the contractor.

In the context of DoD data analytics, "VACP" most closely refers to:

✓ A. Variance at Completion Percent: a measure of projected cost overrun relative to BAC

B. Value Added Cost Performance: a profit metric

C. Vendor Acquisition Compliance Protocol: a contractor reporting standard

D. Validated Acquisition Cost Projection: used in CAPE independent estimates

WHY

VAC (Variance at Completion) = $BAC - EAC$. It represents the projected cost overrun or underrun at program completion. Expressed as a percentage (VAC%), it gives leadership a quick view of how far off budget the program is projected to finish. A negative VAC% means projected overrun.